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| This google sheet is created by <b>Dr Mostafa Saad Ibrahim</b> . Overall ~950 problems for newcomers to problem solving.                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                   |                                                     |                                                                                      |                                           |
| Problem Solving Sheet                                                                                                                          | <a href="mailto:mostafa.saad.fci@gmail.com">mostafa.saad.fci@gmail.com</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <a href="#">Ask.fm</a>            | <a href="#">Site / More Contacts</a>                |                                                                                      |                                           |
|                                                                                                                                                | <a href="#">Video Introducing roadmap (Arabic) - to min 18 ONLY</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                   | <a href="#">Video explaining the sheet</a>          |                                                                                      |                                           |
|                                                                                                                                                | Currentet Version <b>V7.0</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                   | <a href="#">Latest Version</a>                      |                                                                                      |                                           |
|                                                                                                                                                | <ul style="list-style-type: none"><li>- Complete and consistent roadmap for newcomers: What to solve &amp; algorithms to learn in order</li><li>- In the bottom row, there are different sheet pages such as Faq, Topics, CF-C2</li><li>- CF-C1, C2 are (Codeforces Div2 C problems (or similar level from other OJs), but from easy to hard). Same for CF-D1, D2, D3</li><li>- Covering most of topics needed up to codeforces Div2-D</li><li>- Problems of scales 1 - 5.5 / 10 + Few harder ones</li><li>- Problems increase in difficulty per topic with intermediate easy/medium problems + ad-hoc problems</li><li>- Speed problems to maintain speed goals</li><li>- A lot of recorded videos for problems solutions, especially for the entry levels (Arabic)</li><li>- Several students followed its order and managed to solve by themselves 95% of it (up to his current sheet page)</li></ul> <p>- You can train in one of the following ways:</p> <ul style="list-style-type: none"><li>- A) <b>Blind-Order training style</b><ul style="list-style-type: none"><li>- Problems are distributed in sheets CF-A, CF-B, CF-C1, ....CF-D3</li><li>- This one is a roadmap. It targets learning the knowledge/skills in a consistent and balanced way</li><li>- Every sheet page is on average harder than the previous sheet page</li><li>- This is my <b>recommended</b> way, though most camps/training-approaches don't use this style</li></ul></li><li>- B) <b>Topics-Based training style</b><ul style="list-style-type: none"><li>- See sheet page (Topics1). It has the same sheet problems (CF-A to CF-D3) ordered by category and level, around 950 problems</li><li>- <b>Ideas Quality column:</b> P5 (important), P4(very interesting), P3(interesting), P2(good), P1(ok), Empty (normal)</li><li>- Say your level is 6/10, and solved a problem of level 3 with P5, you will find it a normal one. So notice, it is subjective to your level/background</li><li>- You can train using Blind-Order, and use Topics page as guide to skip some problems</li><li>- Many guys/training camps are fan of this topics-based way.</li><li>- You need to be careful with such style as it may corrupt your training quality, e.g. due to your bias</li><li>- Advantage: Mastering the algorithm till solving some hard problems in a short time</li><li>- Disadvantage: Discovering the algorithm behind the problem is an important skill. Given that you know the topic, you lose a good space to improve this skill</li><li>- Disadvantage: Being in the mode of specific algorithm lets you solve many of it easier. However, when solving in real contests, your mind is not so active on the specific topic</li><li>- It is still a good training roadmap. Actually used by most of people I think.</li></ul></li></ul> <p>- See Topics2 page (for extra topics/problems in specific cases)</p> |                                   |                                                     |                                                                                      |                                           |
| What is this Sheet?                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                   |                                                     |                                                                                      |                                           |
| Advantages of this Sheet?                                                                                                                      | <ul style="list-style-type: none"><li>- To be a strong contestant, one has to take care of a number of quality and quantity factors<br/>==&gt; This roadmap does its best to satisfy that</li><li>- Typical issues in our Arabian region: Guys with 700-1000 solved problems and still weak!</li><li>- Why?<ul style="list-style-type: none"><li>A) No specific roadmap or keep switching between them</li><li>B) Training while knowing problem category / level</li><li>C) Focus on specific online judge</li></ul></li><li>- Again, this sheet solves these issues</li><li>- Allows you to write down your statistics to learn from them (e.g. you consume much time in debugging)</li><li>- Continuous refining based on feedback</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                   |                                                     |                                                                                      |                                           |
| Your Sheet <b>COPY</b>                                                                                                                         | <p>This is a personal Google sheet for you [<b>Make a copy</b> from file MENU] to have sets of problems to solve coupled with algorithms to learn</p> <ul style="list-style-type: none"><li>- Don't download the sheet, Use it online</li><li>- Can't edit it? Because it is <b>read-only</b>. Read below notes.</li><li>- Just make a copy to your google driver</li><li>- Then work over it online. Following are the details</li></ul> <ul style="list-style-type: none"><li>- Login to ur google Gmail</li><li>- Go to my sheet</li><li>- In the sheet click on 'file' menu</li><li>- select Make copy</li><li>- it will create copy for u</li><li>- <b>RENAME</b> it to Junior Sheet &lt;Your name&gt;</li><li>- Now the copied sheet is opened for you (or go inside ur Google drive and you will find it)</li></ul> <p>NOTE: If u did so and still read-only format, then you are again opening my sheet (e.g. with old name), NOT your copy</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                   |                                                     |                                                                                      |                                           |
| For Whom?                                                                                                                                      | A junior is anyone who doesn't master solving codeforces Div2-D.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                   |                                                     |                                                                                      |                                           |
| Prerequisites?                                                                                                                                 | Basic Programming skills such in series                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                   | <a href="#">C++ Programming</a>                     | + STL + Debugging Skills                                                             | <a href="#">C++ is highly recommended</a> |
|                                                                                                                                                | If you find my sheet is hard, Finish <b>Assiut</b> University provides an easier starting roadmap. Finish it first                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                   | <a href="#">Novice RoadMap</a>                      | <a href="#">Online Judge</a>                                                         |                                           |
|                                                                                                                                                | Know about our community and what is programming competitions ==>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                   | <a href="#">Watch these videos for more details</a> |                                                                                      |                                           |
|                                                                                                                                                | Code with any language but preferred C++ or Java. For <b>Java</b> :                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                   | <a href="#">Solver to be Channel</a>                | <a href="#">Code El Masry Channel</a>                                                |                                           |
| Training Style?                                                                                                                                | You can train alone, but highly advised to find partner(s) to work with to encourage each other.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                   |                                                     |                                                                                      |                                           |
| Skills Goals                                                                                                                                   | Moving from Junior Level to Semi-Senior Level: A one who do pretty well in CF-Div2 A, B, C, D and similar levels (e.g. TC-Div2-1000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                   |                                                     |                                                                                      |                                           |
| Knowledge Goals                                                                                                                                | Understand and build fair knowledge in some algorithms in Number Theory, Dynamic Programming, Greedy, Graph Theory and Search                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                   |                                                     |                                                                                      |                                           |
| Sheets                                                                                                                                         | Sheet pages are mainly for Codeforces Div2 A, B, C, D + Problems on knowldge topics (Mainly from UVA, SPOJ)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                   |                                                     |                                                                                      |                                           |
|                                                                                                                                                | Each sheet has some sets, each set is ~10-15 problems....The top sets are mandatory....The below sets (after line mark) are optional                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                   |                                                     |                                                                                      |                                           |
|                                                                                                                                                | If you did well in the mandatory sets, move to next sheet...otherwise you still need training on similar level...then solve the optional problems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                   |                                                     |                                                                                      |                                           |
|                                                                                                                                                | Please watch the videos in order, solve UVA/SPOJ problems in order. Don't skip them.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                   |                                                     |                                                                                      |                                           |
|                                                                                                                                                | In some columns, some time recordings. This helps you to know how much time you take per a problem...use that to recognize your problems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                   |                                                     |                                                                                      |                                           |
|                                                                                                                                                | In the level column give an estimate to the problem level from 1-2 (easy), 3-4 (medium), 5-6(hard), 7-8 (had to read editorials), 9-10 (can't solve)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                   |                                                     |                                                                                      |                                           |
|                                                                                                                                                | In the comments column..write comments for hard problems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                   |                                                     |                                                                                      |                                           |
|                                                                                                                                                | <a href="#">Put problem Status</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <a href="#">AC (for Accepted)</a> | <a href="#">CS (can't solve)</a>                    | <a href="#">Other values: WA (wrong answer), TLE (time limit exceeded), RTE, MLE</a> |                                           |
| If you solved a problem before, put ACX instead of AC. Don't resolve                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                   |                                                     |                                                                                      |                                           |
| Don't let a problem consumes more than 2-3 hours. If can't solve it, see editorials/solutions. If still can't solve it, just leave it for now. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                   |                                                     |                                                                                      |                                           |
| Don't compare yourself with others. People vary in their progress                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                   |                                                     |                                                                                      |                                           |

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|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|--|--|--|
| Problem Level Column        | Use the following guide to assign a proper problem level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                              |                                                    |  |  |  |
|                             | Div2-A => 1 - 2<br>Div2-B => 1.5 - 3<br>Div2-C => 3 - 5.5<br>Div2-D => 5 - 6.5<br>Div2-E => 6 - 7.5<br>Div1-D => 7 - 8.5<br>Div1-E => 8 - 9.75<br><br>In other words, most of the time, one shouldn't assign Div2-A problem level such as 5. But it can be: 1, 1.5, 2. Very few might be 2.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                              |                                                    |  |  |  |
| Notations                   | CF136-D2-A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CF (codeforces), D2 (Division 2), (136, A) is the problem URL. Note this is not Round 136 ... it is Round 97                                 |                                                    |  |  |  |
|                             | SRM150-D2-1000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | SRM 150 (Topcoder), D2 (Division 2), 1000 (3rd problem)                                                                                      |                                                    |  |  |  |
| Problems Colors             | For Topcoder:<br><br>If using the applet arena, then we normally use SRM number (the old way)<br>However, if using the web arena, you need first the problem name itself! (the new way - <a href="https://arena.topcoder.com/">https://arena.topcoder.com/</a> )<br><br>The fastest way to get problem name is from the editorial if exist<br><br><a href="https://apps.topcoder.com/wiki/display/tc/Algorithm+Problem+Set+Analysis">https://apps.topcoder.com/wiki/display/tc/Algorithm+Problem+Set+Analysis</a> [LOGIN here first]<br><a href="https://www.topcoder.com/tc?d1=match_editorials&amp;d2=archive&amp;module=Static">https://www.topcoder.com/tc?d1=match_editorials&amp;d2=archive&amp;module=Static</a><br><a href="https://www.topcoder.com/blog/tag/srm-editorials/">https://www.topcoder.com/blog/tag/srm-editorials/</a> [for recent SRMs]<br><br>How to see contest submissions: <a href="https://www.quora.com/Where-can-I-find-the-solutions-to-the-problems-in-TopCoder">https://www.quora.com/Where-can-I-find-the-solutions-to-the-problems-in-TopCoder</a><br><br>Otherwise from match archive: <a href="https://www.topcoder.com/tc?module=MatchList">https://www.topcoder.com/tc?module=MatchList</a><br><br>For notes and tricks for using Java arena applet: See <a href="https://goo.gl/Q43tRL">https://goo.gl/Q43tRL</a> . We can download code with all cases on local machine and code normally.<br><br>Topcoder nowadays maybe slow. So wait for 5 min for a problem to open. If did not work, try in another day<br><a href="https://codeforces.com/blog/entry/61252">https://codeforces.com/blog/entry/61252</a> |                                                                                                                                              |                                                    |  |  |  |
|                             | CF483-D2-A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | White for a problem from codeforces                                                                                                          |                                                    |  |  |  |
|                             | UVA 10242                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Basic (if possible) Knowledge problem on the just watched videos                                                                             |                                                    |  |  |  |
|                             | SPOJ CDOWN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | A knowledge problem on topic you watched before, will be harder than basic problems                                                          |                                                    |  |  |  |
|                             | CF118-D2-B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Problem of easier level than current sheet page level to enhance multiple training levels in same time instead of 1 level training           |                                                    |  |  |  |
|                             | Moving faster                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Do I have to solve every problem? For Div2 (A, B, C1) => No. If you can move faster, do it. For Non CF problems (E.g. UVA), please solve all |                                                    |  |  |  |
|                             | Others Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | If you solved a problem, please see some other accepted solutions in codeforces. You don't need to watch my linked videos unless can't solve |                                                    |  |  |  |
|                             | External Resources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Awesome Competitive Programming                                                                                                              | Many awesome links - very helpful for English guys |  |  |  |
|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Ahmed Elsayghir Traininig                                                                                                                    | Ahmed is senior from GUC                           |  |  |  |
|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | A2oj Ladders                                                                                                                                 | Don't like my sheet? Go with Ahmed Aly Ladders     |  |  |  |
| Prgramming Ahmed M sayd     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Arabic Programming Playlist                                                                                                                  |                                                    |  |  |  |
| Programming Mohamed desouky |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Arabic Programming Playlist                                                                                                                  |                                                    |  |  |  |
| More Resources              | Each video is part of a playlist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                              |                                                    |  |  |  |
| History                     | V1: initial release                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                              |                                                    |  |  |  |
|                             | V2: Vidoes updates. Sheet P2A: Little problems replaced + reordering. P2B, P2C, P2D merged in P2B. P3A and P3B: new knowledge sheets                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                              |                                                    |  |  |  |
|                             | V3: Added problem names. P3A, P3B split over 3 sheets, reordreded to be more incremental rather than random                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                              |                                                    |  |  |  |
|                             | V4: ( <a href="https://docs.google.com/spreadsheets/d/12YI86X40xGtid9t1dUHkK6urrqh6nTaPEvOKBkAbAgU/edit?usp=sharing">https://docs.google.com/spreadsheets/d/12YI86X40xGtid9t1dUHkK6urrqh6nTaPEvOKBkAbAgU/edit?usp=sharing</a> )<br>Solving many knowledge concerns:<br>1- Discarded rare topics (and their problems): ~20 videos.<br>2- Adding Easy problems after each video. You don't have to search by yourself anymore<br>3- Distributing many of the knowledge problems inside the the main sheets instead of delaying them to the last sheets.<br>Other concerns:<br>4- More smooth transitions from a sheet to another<br>5- Utilizing the new many problems added by CF since initial sheet creation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                              |                                                    |  |  |  |
|                             | V5<br>- Added Video Solutions to some existing problems<br>- Added Easier DP problems after its Intro videos<br>- Added new topics: Tree Diameter, Isomorphism, DP (bitmasks, games, probability), Max Flow, SCC, Segment Tree, 2 pointers, Trie, KMP, Geometry Polygons<br>- Added problems for old categories to balance the available problem levels per category.<br>- Added 3 sheets for Div2-D (contains the old Misc sheet problems)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                              |                                                    |  |  |  |
|                             | Note: If you were using version 4.X, then the major change for you is replacing "Misc" sheet with the 3 Div2-D sheets. If wanna migrate:<br>- Then Remove Misc sheet<br>- Click on the arrow for Div2-D sheets, and make copy for your sheet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                              |                                                    |  |  |  |
|                             | V6:<br>- Added 3 columns to the sheet: debug time, category and by yourself columns<br>- Each sheet is enhanced with problems from the lower sheet (shifted from it). The purpose is to mix levels per sheet, hence allow multiple training levels in same time (hard vs speed concern). See the new added color<br>- Added probability/expectations English videos/problems<br>- Added Topological sort problems<br>- Solution editorials linked to many non-CF problems / more videos in Dlv2-A/Div2-B<br>- Add many problems where my trainees marked as interesting problems. Removed some problems that I think not that interesting or its ideas covered by other problems (subjective). I am working on sheets with a simple, but hard to do idea: Most of the problems seems for the trainer novel in idea with less repeated ideas, hence learning a lot while solving much less.<br>- Add topics based training style sheet page                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                              |                                                    |  |  |  |
|                             | V7:<br>- Added Topics2 (See notes there) - not intended for juniors                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                              |                                                    |  |  |  |
|                             | ====<br>If you are working in some sheet, find the convenient point to switch. E.g. if you are in middle of sheet, finish it and move to new sheet page from next one. Say you are in middle C1. Finish it first. Then remove C2, D1, D2, D3. In the new version, click a sheet page and select Copy To, then copy to your sheet. In other words, <b>migration</b> should be 5-min process. If need more, you are doing it wrongly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                              |                                                    |  |  |  |

|                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Thanks</b> for all guys who sent sheet feedback: Mariam Alshereef, Magdy Hassan, Ahmed Yasser, Ahmed Elsayed Awad, Mohamed Nasser, Mostafa Ali Mansour, Aya elymany, Ayyad shenouda, Others. |
| Special Thanks for Coach Alhussain Aly for his continuous help                                                                                                                                  |
| Special Thanks for All volunteers in videos recording / Editorials writing                                                                                                                      |

**Q) What is the sheet requirements? Should I study algorithms and Data structures?**

- ONLY programming skills (e.g. Programming 1 level). It is highly advised to implement 2-3 projects
- NO for OOP
- NO for datastructures, but learn STL (or Collections in Java/C#). It helps alot
- NO for algorithms, the sheet will teach you that in a smooth way
- For C++ guys (and others as guide) - first 18 videos here: <https://www.youtube.com/playlist?list=PLPt2dINI2MIZPFq6HyUB1Uhx1UDnZMS>

**Q) How much time do I need to finish the sheet?**

- Answer varies from one to another.
- Some trainees are fast through whole sheet
- Some trainees are slow through whole sheet
- Some trainees are fast in early pages, but slow at the end
- Some trainees are slow in early pages, but fast at the end
- The sheet has ~900 problems. Around 60 videos. Problems targets average guys. However, you are encouraged to skip problems whenever you could. I expect many guys could skip 20%-30% of the mandatory problems. Make use of the Topics page.

215 problems of level <= 2.5 (avg 20 min per problem)  
93 problems of level <= 3.5 (avg 30 min per problem)  
270 problems of level <= 4.5 (avg 40 min per problem)  
178 problems of level <= 5.25 (avg 60 min per problem)  
127 problems of level <= 5.75 (avg 75 min per problem)  
53 problems of level > 5.75 (avg 90 min per problem)

$215 \times 20 + 93 \times 30 + 270 \times 40 + 178 \times 60 + 127 \times 75 + 53 \times 90 = \sim 700$  hours (say max 900 hours)

If you trained in the summer vacation seriously for 2 month (e.g. 10 hours \* 30 days \* 2 month = 600 hours) + the remaining of the year effort, you could solve the whole sheet smoothly and move to Div2-E level goal

- <https://ask.fm/mostafasaad87/answers/144907000290> [adjust to whatever fits with you]

**Q) When should I give up and check the editorials and solutions?**

<https://ask.fm/mostafasaad87/answers/144907000290>

**Q) Got WA, should I check directly the test cases?**

- No, remember in a real contest you only know your problem status (WA, TLE, ...etc)
- Struggle to find the wrong case by yourself. At least 15-30 minutes.
- Don't keep trying longer, just check the test cases
- If you can write a brute force solution for your problem, write a stress test: Generate random cases and compare the optimal algorithm with the brute force case

**Q) What is the debug time?**

- Once you finish coding and start testing, you verify if the program is working as expected or not
- If not, there are bugs that you need to find to make the program behave as expected. From this moment till getting the program AC = debugging time
- People could debug using 'print statements'. A better way using a debugger
- Check out these 4 videos: <https://www.youtube.com/watch?v=DlbQwQEiDW0&t=0s&index=35&list=PLPt2dINI2MIZPFq6HyUB1Uhx1UDnZMS>

**Q) Should I solve every problem?**

- Generally, preferred, but If you think certain level is easy (e.g. solve it within 15 minutes), then jump a block and so on
- This jumping might be for codeforces problems only

**Q) Just started in Div2-A, could I finish its codeforces problems first, then solve the UVA/Colored problems?**

- Many juniors find UVA problems in Div2-A hard, I understand
- Yes, almost same for Div2-B. But don't do that in next sheets as order might matter, because all of such knowledge are mainly preparations for hard Div2-B or Div2-C
- However, following the order is a much better idea
- Similarly, one could finish All Div2-A/Div2-B codeforces problems, then solve their colored problems. Again, this is not the best way.

**Q) Is using C# ok?**

- Generally yes, but you won't be able to submit in UVA judge, as C# is not supported
- For such problems, write your code, but heavily test it. You may download an internet code and evaluate the test case on both
- On the other hand, learning Basic C++ + STL is not hard for C#/Java guys
- C++/Java/Python are official in UVA
- Codeforces allows more such as Javascript

**Q) When I watch a video, should I solve the problems in its info section?**

- No, sheet has subset of these problems already in specific order
- Sheet is self-contained

**Q) I watched the video, but it is hard, any tips?**

- Algorithms are hard, learn to struggle
- Watch the video 2-3 times, try to rewrite its code by yourself
- Still can't get it? Google for more materials from the web (ppt/pdf/videos) and try to learn
- In worst case, leave it for now and return to it later

**Q) How does your sheet prepare for ECPC/ACPC?**

- The sheet prepares you to reach level 5-5.5/10 in several categories
- If a team of 3 members solved the whole sheet, they may rank in the top 15 in the contest

But let's go in details. Individual success in contests depends on several factors. Let me state some of them.

- Solving many problems of good quality
- Improving your different skills (reading, thinking, coding and accuracy).
- 2 persons could solve in training the same problem. One got it in 20 min first submission, and the 2nd needed 90 minutes due to 60 minutes debugging.
- Healthy training: Regular / good times for training (e.g. morning) / weekly contests / reading other codes / collaboration with others / etc
- Stress management during contests
- Emotions management when fail in solving or feel performance is not improving enough
- Avoiding Psychological issues: Comparing to others, Negative feelings, Your image, Regretting training time

Moving toward a team contest, you need more concerns:

- Serious team members. If only one active member, they may end up in bad performance. So EACH team member need to finish the sheet individually + weekly contests
- Tolerating team mistakes during the contest
- PC management
- Suitable strategy + several team contests to tune it

As you may notice, there are MANY factors for success.

- This sheet provides you with high quality problems and good topics distribution + way to record your stats to know your weak points
- However, there are many concerns that YOU have to tackle by yourself and your team members

- Finish up to CF-C2 sheet, then study from the "Cracking the Coding Interview: 150 Programming Questions and Solutions" book
- Also watch: <https://www.youtube.com/watch?v=39vqarATPyM>

**Q) How different is your sheet versus Ahmed Aly Ladders?**

- Ladder problems are selected automatically, no personal investigation for the actual benefit/need from the problem
- Mine is mixed between automated and manual.
- At the current moments, many of my trainees and students feedback, I am aware of the problem level and its category.
- I updated the sheet many times because of the received feedback
- My sheet involves the algorithms videos to learn, in order, while you grow up.
- I selected videos to prepare you as soon as possible for Div2-C/Div2-D where many algorithms starts to appear
- It is a sheet..ready for you to record your times, notes...etc...this help to improve yourself
- It is not blocking style. If you can't solve problem, just leave it and move to other one. In ladders, you see next problem when solve current one (or do workarounds)

#### Q) How did you **select problems** for the sheet?

- Long story, many versions were there, from a version to another improvements were applied
- Codeforces problems where rated based on this CF tool: [http://codeforces.com/blog/entry/46304?mobile=true#\\_=\\_](http://codeforces.com/blog/entry/46304?mobile=true#_=_)
- Any rating is just an estimation. I found this one a pretty reasonable measure
- The videos are selected such that when comes to Div2-C/, you are ready
- Manual selections and investigations for non-CF problems to be used in the sheet
- Lots of manual efforts and investigations and feedback processing

#### Q) what is the next step **after finishing** your sheet ?

- Joining directly my ICPC semi-seniors supervision, **BUT**
- Email me with your online sheet copy link and it **must** have
- Each row should have: code link, time details, problem level, category and comment per a problem
- I will review and decide
- Side note: If you started in Div2-C1 and solved first 15 problems, you can share the sheet with me to follow your updates

#### Q) can't **access** the sheet in **edit** mode?

- Don't download the sheet, Work over it online "better"
- Can't edit it? Because it is read-only. Read below notes.
- Just make a copy to your google driver
- Then work over it online. Following are the details
- Login to ur google Gmail
- Go to my sheet
- In the sheet click on 'file' menu
- select Make copy
- it will create copy for u
- RENAME it to Junior Training Sheet
- Now the copied sheet is opened for you (or go inside ur Google drive and you will find it)

NOTE: If u did so and still read-only format, then you are again opening my sheet (e.g. with old name), NOT your copy

#### Q) What to write in the **category** column?

The algorithm used to solve a problem. In Div2-A, this might be:

- This column is for the algorithm you used during solving. Usually, new guys in CF-A are confused. If so, leave it CF-A and start to write in CF-B
- The more you go in the sheet you will learn algorithms (e.g. Binary search, DP, DFS, etc). Then this what you write in level column
- The problem that has no algorithm but a specific idea called ad-hoc, This is the case for most of CF-A and less later
- Implementation: Means the problem request is almost direct, just code it
- Brute Force: Means instead of finding elegant solution, try all possible solutions (e.g. 3 nested loops) and select the solution
- Ad-hoc: Just per-problem thinking in a special way/analysis on how to solve the problem
- Please watch from this minute: <https://youtu.be/DZ6YTtLCE8?t=839>

#### Q) Are problems really sorted based on **easiness**? I don't feel so.

They are sorted by easiness already. But, whatever order, anyone will find some are easy and some are hard in some order.

That is, no one can give you a list that every problem for YOU is easier than the next problem.

In other words, If we gave 100 problems to 10 students of same level to solve and told them rank from easy to hard, they will rank them differently.

So, questions ordered by people average. The promise is, the problems will be within your range to solve.

#### Q) What are these problems **colors**?

See "Problems Colors" notes in info page

#### Q) Are the problems **sorted**?

Yes, but this is tricky as sorting is subjective.

That is imagine 10 prblms given for 100 people to order based on its level, people will arrange in different ways based on their experience

So if you felt they are not sorted, just keep going

#### Q) Why problem-solving is that **important**?

See the first 2 videos here: <https://www.youtube.com/playlist?list=PLPt2dINi2MIaNCuJ070HIAO8JWYBcafuyG>

#### Q) I **feel bored** when solving problems compare to doing projects?

<https://ask.fm/mostafasaad87/answers/145333554402>

#### Q) I would like to **freeze** my study for 1-2 years to be good in problem-solving?

I never liked that. Graduate on time. In your free times and vacations do more problem solving

Relevant: <https://ask.fm/mostafasaad87/answers/145151822818>

#### Q) Topics based-training vs Blind Order

In topics training, we study a topic, then solve a lot of problems over it.

Advantages:

- Mastering the algorithm till solving some hard problems in short time

Disadvantages:

- Discovering the algorithm behind the problem is an important skill. Given that you know the topic, you lose a good space to improve this skill
- Being in the mode of specific algorithm lets you solve many of it easier. However, when solving in real contests, your mind is not so active on specific topic

In my sheets - Blind style:

- You solve 3-5 per topic. Then you have to discover the other problems by yourself. So you train to avoid the missing 2 points

Claim:

- Although topics training let guys be so good early, they level stuck early and they don't improve. Seems to me, topics training is an important factor in doing so. Meanwhile, if you just target to be good in Div2D level in shorter time and no interest in further competitions achievements, you may go topics based.

#### Q) Who Finished my sheet? Their levels?

<https://ask.fm/mostafasaad87/answers/150802497762>

#### Q) How to share my sheet progress with you?

<https://ask.fm/mostafasaad87/answers/148552940002>

Q) What is after the sheet?

- There are 2 other levels, each has around 1000 problems. Semi-senior level and seniors level
- Generally speaking, the region stars will solve a lot of problems, e.g. 2000-3000 problems with many of them of hard level
- Whoever finish the sheet, I join him in my supervision for the next levels

| Problem Name     | Problem Code                  | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments About your approach                                                                                                                                                                                                                           |
|------------------|-------------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | AC Averages =>                | 3      | 2.3          | 5               | 13               | 15             | 18            | 50            | 2                 | 2            | 2        | 2                                                                                                                                                                                                                                                               |
| Sample Name1     | Sample Link1                  | AC     | 5            | 4               | 8                | 6              | 32            | 50            | 2                 | Yes          | Math     | Solution in mind is O(n^2): for each pair of points, get the equation of the straight line linking between them, and add 2 to its count. Print the count of the line having the max count. Problem is how to hash a line equation (coefficients are double).    |
| Sample Name2     | Sample Link2                  | AC     | 1            | 5               | 10               | 35             | 20            | 70            | 2                 | No           | Impl     | <b>My Performance Notes:</b> This is so bad performance. Needed many submissions per problem. Always submit as if you are in real contest. Submit to AC, not to see if we will pass or not. <b>Target AC from 1st</b> submission. Think more before submission. |
| Sample Name3     | Sample Link3                  | AC     | 1            | 5               | 20               | 4              | 1             | 30            | 2                 | Yes          | Graph    | Please always write and study your timings.                                                                                                                                                                                                                     |
| Sample Name4     | Sample Link4                  | WA     | 5            | 4               | 25               | 20             | 2             | 51            | 7                 | Hint         | Math     | <b>My Performance Notes:</b> This is so good performance. 1st submission. Thinking is the higher. Code/Debug is so low. By time, thinking column will be improved.                                                                                              |
| Sample Name5     | Sample Link5                  | CS     | 6            | 5               | 30               | 25             | 31            | 91            | 9                 |              |          | Other Status values: AC, WA, CS, TLE, MLE, RTE, ... These values and comments are just examples. Just remove/ignore them.                                                                                                                                       |
|                  |                               |        |              |                 |                  |                |               | 0             |                   |              |          | Want c++ solution for UVA 408? Google with: UVA 408 filetype:cpp                                                                                                                                                                                                |
|                  |                               |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Approaching Problem Statement</a>                                                                                                                                                                                                           |
|                  |                               |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Thinking - On papers Not on PC</a>                                                                                                                                                                                                          |
| Vanya and Feni   | <a href="#">CF677-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">C++ Solution Example</a>                                                                                                                                                                                                                            |
| Anton and Dani   | <a href="#">CF734-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">This is from Round 379. Here is the editorial</a>                                                                                                                                                                                                   |
|                  |                               |        |              |                 |                  |                |               | 0             |                   |              |          | <i>You shouldn't watch a solution video unless you can't solve it by yourself and don't get it from editorial/code. Videos are there just to for extra help.</i>                                                                                                |
|                  |                               |        |              |                 |                  |                |               | 0             |                   |              |          | In the first 20 problems, don't think more than 20 minutes. After that see the solutions.                                                                                                                                                                       |
|                  |                               |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                                                                                 |
| Bear and Big Bl  | <a href="#">CF791-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Youssef El Ghareeb</a>                                                                                                                                                                                                         |
| Team             | <a href="#">CF231-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Youssef Ali</a>                                                                                                                                                                                                                |
| Beautiful Matrix | <a href="#">CF263-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Samed Hajaj</a>                                                                                                                                                                                                                |
| Gravity Flip     | <a href="#">CF405-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng John Gama</a>                                                                                                                                                                                                                  |
| Petya and Strin  | <a href="#">CF112-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Boy or Girl      | <a href="#">CF236-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Word             | <a href="#">CF59-D2-A</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Magnets          | <a href="#">CF344-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Sereja and Dim   | <a href="#">CF381-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Stones on the T  | <a href="#">CF266-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ahmead Raafat (Python)</a>                                                                                                                                                                                                     |
| Police Recruits  | <a href="#">CF427-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ahmead Raafat (Python)</a>                                                                                                                                                                                                     |
| Black Square     | <a href="#">CF431-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ahmead Raafat (Python)</a>                                                                                                                                                                                                     |
| Night at the Mu  | <a href="#">CF731-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Yahia Ashraf</a>                                                                                                                                                                                                               |
| Games            | <a href="#">CF268-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Yahia Ashraf</a>                                                                                                                                                                                                               |
|                  |                               |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                                                                                 |
|                  |                               |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Measuring Algorithms Performance - 1</a>                                                                                                                                                                                                    |
|                  |                               |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Elementary Math - Introduction</a>                                                                                                                                                                                                          |
| Buy a Shovel     | <a href="#">CF732-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Yahia Ashraf</a>                                                                                                                                                                                                               |
| Is your horse sh | <a href="#">CF228-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ahmead Raafat (Python)</a>                                                                                                                                                                                                     |
| Colorful Stones  | <a href="#">CF265-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ahmead Raafat (Python)</a>                                                                                                                                                                                                     |
| Die Roll         | <a href="#">CF9-D2-A</a>      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                                                                                                                                                                                                         |
| Shaass and Osi   | <a href="#">CF294-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                                                                                                                                                                                                |
| Juicer           | <a href="#">CF709-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Carrot Cakes     | <a href="#">CF799-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Anton and Lette  | <a href="#">CF443-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Way Too Long     | <a href="#">CF71-D2-A</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Free Ice Cream   | <a href="#">CF686-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Helpful Maths    | <a href="#">CF339-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Team Olympiad    | <a href="#">CF490-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                                                                                                                                                                                                         |
| New Password     | <a href="#">CF770-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                                                                                 |
|                  |                               |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                                                                                 |
|                  |                               |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Number Theory - Modular Arithmetic</a>                                                                                                                                                                                                      |
|                  |                               |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Combinatorics - Counting Principles</a>                                                                                                                                                                                                     |
| Light, more ligh | <a href="#">UVA 10110</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Amr Saud</a>                                                                                                                                                                                                                   |
| Product          | <a href="#">UVA 10106</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Youssef El Ghareeb, Don't solve using big integer</a>                                                                                                                                                                          |
| Uniform Genera   | <a href="#">UVA 408</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Yahia Ashraf</a>                                                                                                                                                                                                               |
| Black and white  | <a href="#">UVA 11231</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Amr Saud</a>                                                                                                                                                                                                                   |
|                  | <a href="#">SPOJ EASYMATH</a> |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                                                                                                                             |
| Electricity      | <a href="#">UVA 12148</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Learn Calender Leap Year</a>                                                                                                                                                                                                                        |
|                  |                               |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                                                                                 |
| Presents         | <a href="#">CF136-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ahmed Rafaat (Python)</a>                                                                                                                                                                                                      |
| Lineland Mail    | <a href="#">CF567-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ahmed Rafaat (Python)</a>                                                                                                                                                                                                      |
| Mahmoud and L    | <a href="#">CF766-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Snacktower       | <a href="#">CF767-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                                                                                 |
| Oath of the Nigl | <a href="#">CF768-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Pangram          | <a href="#">CF520-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Twins            | <a href="#">CF160-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Keyboard         | <a href="#">CF474-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
|                  |                               |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                                                                                 |
|                  |                               |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Graph Theory - Intro</a>                                                                                                                                                                                                                    |
|                  |                               |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Graph Theory - DFS</a>                                                                                                                                                                                                                      |
| The Seasonal V   | <a href="#">UVA 352</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Nasser</a>                                                                                                                                                                                                             |
| Marcus           | <a href="#">UVA 10452</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ayman Salah</a>                                                                                                                                                                                                                |
| Battleships      | <a href="#">UVA 11953</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Aya Elymany</a>                                                                                                                                                                                                                |
|                  |                               |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Read definition of: Bipartite graph</a>                                                                                                                                                                                                             |
| Forming Teams    | <a href="#">CF216-D2-B</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                                                                                                                                                                                                |
| Hierarchy        | <a href="#">SPOJ MAKETREK</a> |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Yahia Ashraf</a>                                                                                                                                                                                                               |
| Ordering Tasks   | <a href="#">UVA 10305</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Yahia Ashraf</a>                                                                                                                                                                                                               |
|                  |                               |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                                                                                 |
| Even Odds        | <a href="#">CF318-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                                                                                                                                                                                                         |
| I Wanna Be the   | <a href="#">CF469-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Is it rated?     | <a href="#">CF807-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Olesya and Roc   | <a href="#">CF584-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Football         | <a href="#">CF43-D2-A</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Belal Abdounasser (Python)</a>                                                                                                                                                                                                 |
| Brain's Photos   | <a href="#">CF707-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Dubstep          | <a href="#">CF208-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Valera and X     | <a href="#">CF404-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Arpa's hard exa  | <a href="#">CF742-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Calculating Fun  | <a href="#">CF486-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Theatre Square   | <a href="#">CF1-D12-A</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Anton and Poly   | <a href="#">CF785-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |
| Panoramix's Prr  | <a href="#">CF80-D2-A</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                                                                            |

| Problem Name                              | Problem Code                 | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments<br>About your approach                                                                                                                   |
|-------------------------------------------|------------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | AC Averages =>               | 3      | 2.3          | 5               | 13               | 15             | 18            | 50            | 2                 | 2            | 2        | 2                                                                                                                                                          |
| Counterexample                            | <a href="#">CF483-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                       |
|                                           |                              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                                           |                              |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Computational Geometry - Intro</a>                                                                                                     |
|                                           |                              |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Computational Geometry - Point and Vector</a>                                                                                          |
| Wasted Time                               | <a href="#">CF127-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Points in Figure                          | <a href="#">UVA 476</a>      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Overlapping Rectangles                    | <a href="#">UVA 460</a>      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                                                                                                    |
| Fancy Fence                               | <a href="#">CF270-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Omar Ashraf</a>                                                                                                           |
| Pouring Rain                              | <a href="#">CF667-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Fourth Point !!                           | <a href="#">UVA 10242</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Magdy Hasan</a>                                                                                                           |
|                                           |                              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Good Number                               | <a href="#">CF365-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                                                                                                    |
| Dice Tower                                | <a href="#">CF225-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                                                                                                    |
| Alyona and Nuriya                         | <a href="#">CF682-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng John Gamal</a>                                                                                                            |
| Mountain Scenery                          | <a href="#">CF218-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng John Gamal</a>                                                                                                            |
| Help Vasilisa the Brave                   | <a href="#">CF143-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng John Gamal</a>                                                                                                            |
| Chewbacca and Artoo                       | <a href="#">CF514-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                                                                                                    |
| Ksenia and Pan                            | <a href="#">CF382-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Samed Hajajala</a>                                                                                                        |
| Launch of Collapsing                      | <a href="#">CF699-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Samed Hajajala</a>                                                                                                        |
| Polo the Penguin                          | <a href="#">CF289-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                                                                                           |
| IQ Test                                   | <a href="#">CF287-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                                                                                           |
| Yaroslav and Perpetua                     | <a href="#">CF296-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                                                                                           |
| Snow Footprints                           | <a href="#">CF298-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                                                                                           |
| Raising Bacteria                          | <a href="#">CF579-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ahmed Rafaat (Python)</a>                                                                                                 |
| BowWow and the Game                       | <a href="#">CF1204-D2-A</a>  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                                                                                           |
| Balanced Rating                           | <a href="#">CF1237-D12-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                                           |                              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                                           |                              |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Search Techniques - Binary Search</a>                                                                                                  |
| The Playboy Chimp                         | <a href="#">UVA 10611</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ayman Salah</a>                                                                                                           |
| Pipeline                                  | <a href="#">CF287-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                                                                                           |
| Burning Midnight                          | <a href="#">CF165-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Aggressive cow                            | <a href="#">SPOJ AGGRCOW</a> |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Youssef El Ghareeb</a>                                                                                                    |
|                                           |                              |        |              |                 |                  |                |               | 0             |                   |              |          | Before moving to another sheet, email me with feedback about these problems selection.                                                                     |
|                                           |                              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                                           |                              |        |              |                 |                  |                |               | 0             |                   |              |          | Optional Problems                                                                                                                                          |
|                                           |                              |        |              |                 |                  |                |               | 0             |                   |              |          | You don't have to or encouraged to solve the next problem. If you felt you need so, try some of them. Or Proceed to next and solve in parallel, up to you. |
|                                           |                              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Word Capitalization                       | <a href="#">CF281-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                       |
| Next Round                                | <a href="#">CF158-D12-A</a>  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                       |
| Young Physicist                           | <a href="#">CF69-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                       |
| Bit++                                     | <a href="#">CF282-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                       |
| Case of the Zeros and Ones                | <a href="#">CF556-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                       |
| Translation                               | <a href="#">CF41-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                       |
| String Task                               | <a href="#">CF118-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                       |
| Laptops                                   | <a href="#">CF456-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                       |
| Left-handers, Right-handers and Computers | <a href="#">CF950-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Hossam Yehia</a>                                                                                                          |
| George and Alex                           | <a href="#">CF467-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ahmed Rafaat (Python)</a>                                                                                                 |
| Vasya the Hipster                         | <a href="#">CF581-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Fox And Snake                             | <a href="#">CF510-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| The New Year: A Journey Without Tears     | <a href="#">CF723-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Elephant                                  | <a href="#">CF617-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Greg's Workout                            | <a href="#">CF255-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Ultra-Fast Math                           | <a href="#">CF61-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Little Pony and Big Apple                 | <a href="#">CF454-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| One-dimensional                           | <a href="#">CF721-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Soldier and Bananas                       | <a href="#">CF546-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                                           |                              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Bus to Udayland                           | <a href="#">CF711-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Cookies                                   | <a href="#">CF129-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Second Order Statistics                   | <a href="#">CF22-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Nearly Lucky Numbers                      | <a href="#">CF110-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Playing with Dice                         | <a href="#">CF378-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| A Good Contest                            | <a href="#">CF681-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Beautiful Year                            | <a href="#">CF271-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Far Relative's Birthday                   | <a href="#">CF629-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Mashmokh and Matrix                       | <a href="#">CF415-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Triangular numbers                        | <a href="#">CF47-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                                           |                              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Roma and Luck                             | <a href="#">CF262-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Toy Army                                  | <a href="#">CF84-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Levko and Table                           | <a href="#">CF361-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Cards                                     | <a href="#">CF701-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Wizards' Duel                             | <a href="#">CF591-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Combination Lock                          | <a href="#">CF540-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Summer Camp                               | <a href="#">CF672-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Soft Drinking                             | <a href="#">CF151-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Coder                                     | <a href="#">CF384-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| GukiZ and Contests                        | <a href="#">CF551-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                                           |                              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Circle Line                               | <a href="#">CF278-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Patrick and Sho                           | <a href="#">CF599-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Choosing Team                             | <a href="#">CF432-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Vanya and Cubes                           | <a href="#">CF492-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Insomnia cure                             | <a href="#">CF148-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Cakeminator                               | <a href="#">CF330-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Flag                                      | <a href="#">CF16-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Cupboards                                 | <a href="#">CF248-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Soroban                                   | <a href="#">CF363-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                                           |                              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Amusing Joke                              | <a href="#">CF141-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Lights Out                                | <a href="#">CF275-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Lunch Rush                                | <a href="#">CF276-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Duff and Meat                             | <a href="#">CF588-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Vanya and Carrots                         | <a href="#">CF401-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Squats                                    | <a href="#">CF424-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Arrival of the General                    | <a href="#">CF144-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Sinking Ship                              | <a href="#">CF63-D2-A</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |



| Problem Name      | Problem Code               | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments About your approach |
|-------------------|----------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|---------------------------------------|
|                   | AC Averages =>             | 3      | 2.3          | 5               | 13               | 15             | 18            | 50            | 2                 | 2            | 2        | 2                                     |
| LLPS              | <a href="#">CF202-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Candy Bags        | <a href="#">CF334-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
|                   |                            |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Game With Slic    | <a href="#">CF451-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Vasya and Soci    | <a href="#">CF460-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Dima and Frien    | <a href="#">CF272-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Nicholas and P    | <a href="#">CF678-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Toy Cars          | <a href="#">CF545-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| DZY Loves Has     | <a href="#">CF447-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| HQ9+              | <a href="#">CF133-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Holidays          | <a href="#">CF670-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Dividing Orange   | <a href="#">CF244-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Haiku             | <a href="#">CF78-D2-A</a>  |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
|                   |                            |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| System of Equa    | <a href="#">CF214-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| IQ test           | <a href="#">CF25-D2-A</a>  |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Contest           | <a href="#">CF501-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Restoring Passw   | <a href="#">CF94-D2-A</a>  |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Valera and Plat   | <a href="#">CF369-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Minimum Difficu   | <a href="#">CF496-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Little Elephant s | <a href="#">CF221-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Collecting Beats  | <a href="#">CF373-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Letter            | <a href="#">CF14-D2-A</a>  |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Kefa and First S  | <a href="#">CF580-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
|                   |                            |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Ilya and Bank A   | <a href="#">CF313-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Uncowed Force     | <a href="#">CF604-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Reconnaissance    | <a href="#">CF34-D2-A</a>  |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Lucky Ticket      | <a href="#">CF146-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Chat room         | <a href="#">CF58-D2-A</a>  |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| George and Sle    | <a href="#">CF387-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Ostap and Gras    | <a href="#">CF735-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| The number of i   | <a href="#">CF124-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Table             | <a href="#">CF359-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Tavas and Nafa    | <a href="#">CF535-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
|                   |                            |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Watermelon        | <a href="#">CF4-D2-A</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Let's Watch Foc   | <a href="#">CF195-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Initial Bet       | <a href="#">CF478-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Saitama Destro    | <a href="#">CF608-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Queue on Bus s    | <a href="#">CF435-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Bicycle Chain     | <a href="#">CF215-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Little Elephant s | <a href="#">CF205-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Amr and Music     | <a href="#">CF507-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Marks             | <a href="#">CF152-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Postcards and p   | <a href="#">CF137-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
|                   |                            |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Business trip     | <a href="#">CF149-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Drazil and Date   | <a href="#">CF515-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Multiplication T  | <a href="#">CF577-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Exam              | <a href="#">CF534-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Alena's Schedu    | <a href="#">CF586-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Interview         | <a href="#">CF631-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Lucky Division    | <a href="#">CF122-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Appleman and I    | <a href="#">CF462-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Vasya and Digit   | <a href="#">CF355-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Parallelepiped    | <a href="#">CF224-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
|                   |                            |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Group of Studen   | <a href="#">CF357-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Joysticks         | <a href="#">CF651-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Array             | <a href="#">CF300-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Round House       | <a href="#">CF659-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Lala Land and J   | <a href="#">CF558-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Autocomplete      | <a href="#">CF53-D2-A</a>  |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Digital Counter   | <a href="#">CF495-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Vitaliy and Pie   | <a href="#">CF525-D2-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |
| Life Without Zer  | <a href="#">CF75-D2-A</a>  |        |              |                 |                  |                |               | 0             |                   |              |          |                                       |

|                         | Problem Code                     | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments<br>About your approach                                     |
|-------------------------|----------------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|------------------------------------------------------------------------------|
|                         | AC Averages =>                   | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                                                            |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Thinking - Problem Simplification.</a>                   |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Thinking - Brainstorm - Rank - Approach</a>              |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Study STL (You may study data structures if found hard)</a>      |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Combinatorics - Permutations and Combinations - 1</a>    |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Combinatorics - Permutations and Combinations - 2</a>    |
| Petya and Country       | <a href="#">CF66-D2-B</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                      |
| Bear and Finding C      | <a href="#">CF680-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                      |
| Burglar and Match       | <a href="#">CF16-D2-B</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                      |
| Caisa and Pylons        | <a href="#">CF463-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                      |
| Sum of Digits           | <a href="#">CF102-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                      |
| Coins                   | <a href="#">CF47-D2-B</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Sameh Hajajla</a>                           |
| Effective Approach      | <a href="#">CF227-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Abanob Ashraf</a>                           |
| Easter Eggs             | <a href="#">CF78-D2-B</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Abanob Ashraf</a>                           |
| Decoding                | <a href="#">CF746-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                         |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Training-Secrets of Success</a>                          |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Revise Stack/Queue datastructure concepts. Learn using STL</a>   |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Number Theory - Fib, GCD, LCM, Pow</a>                   |
| Big Mod                 | <a href="#">UVA 374</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                              |
| Combinations            | <a href="#">UVA 369</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                              |
| Pi                      | <a href="#">UVA 412</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Adel</a>                            |
| Adding Reversed Numbers | <a href="#">UVA 713</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | Don't use big integer class. Write simple array computations                 |
| Taxi                    | <a href="#">TIMUS 1607</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | Can you get AC first submission?                                             |
| The Drunk Jailer        | <a href="#">LIVEARCHIVE 2557</a> |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Find a formula</a>                                               |
| Vanya and Lantern       | <a href="#">CF492-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                         |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Prefix Sum</a>                                           |
| Kuriyama Mira's S       | <a href="#">CF433-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                              |
| Fence                   | <a href="#">CF363-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                      |
| President's Office      | <a href="#">CF6-D2-B</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                      |
| Lovely Palindrome       | <a href="#">CF688-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                         |
| Sort the Array          | <a href="#">CF451-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                         |
| Summer sell-off         | <a href="#">CF810-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                         |
| Colorful Field          | <a href="#">CF79-D12-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                         |
| Keyboard                | <a href="#">CF88-D2-B</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                      |
| Mahmoud and a Tri       | <a href="#">CF766-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                         |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                              |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Graph Theory - BFS</a>                                   |
| Tic-Tac-Toe ( I )       | <a href="#">SPOJ TOE1</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ayman Salah</a>                             |
| Tic-Tac-Toe ( II )      | <a href="#">SPOJ TOE2</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Essam AlNaggar</a>                          |
| Knight Moves            | <a href="#">UVA 439</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Magdy Hasan</a>                             |
| King's Path             | <a href="#">CF242-D2-C</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                             |
| Bookworm                | <a href="#">TIMUS 1638</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Can you get AC first submission</a>                              |
|                         | <a href="#">UVA 10461</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                              |
|                         | <a href="#">SPOJ POSTERIN</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                          |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                              |
| Students and Shoe       | <a href="#">CF129-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Abanob Ashraf</a>                           |
| Dreamoon and WiF        | <a href="#">CF476-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Adel</a>                            |
| Chat Online             | <a href="#">CF469-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Adel</a>                            |
| Olympic Medal           | <a href="#">CF215-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ahmed Salah</a>                             |
| Filya and Homework      | <a href="#">CF714-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                      |
| Inna and New Matr       | <a href="#">CF400-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Salah</a>                           |
| Steps                   | <a href="#">CF152-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                      |
| Growing Mushroom        | <a href="#">CF186-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Salah</a>                           |
| Escape                  | <a href="#">CF148-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ahmed Salah</a>                             |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Review - Recursion</a>                                           |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Intro to DP - 1</a>                                      |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Intro to DP - 2</a>                                      |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                              |
| Vacations               | <a href="#">CF699-D2-C</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                              |
| Woodcutters             | <a href="#">CF545-D2-C</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                              |
| Barcode                 | <a href="#">CF225-D2-C</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                             |
| Continents              | <a href="#">UVA 11094</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ayman Salah</a>                             |
| Brownie Points          | <a href="#">UVA 10865</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Magdy Hasan</a>                             |
| Hanoi Tower             | <a href="#">TIMUS 1054</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                          |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                              |
| Roma and Changir        | <a href="#">CF262-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Salah</a>                           |
| Bear and Strings        | <a href="#">CF385-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Salah</a>                           |
| I.O.U.                  | <a href="#">CF376-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Abanob Ashraf</a>                           |
| Jeff and Periods        | <a href="#">CF352-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                      |
| Meeting                 | <a href="#">CF144-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                      |
| Chocolate               | <a href="#">CF617-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Yahia Ashraf</a>                            |
| Easy Number Chal        | <a href="#">CF236-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Yahia Ashraf</a>                            |
| Han Solo and Laze       | <a href="#">CF514-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                              |
| Physics Practical       | <a href="#">CF253-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Salah</a>                           |
| Two Buttons             | <a href="#">CF520-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                         |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                              |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Computational Geometry - Complex Number and 2D Point</a> |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Computational Geometry - Lines and Distances</a>         |
| Intersecting Lines      | <a href="#">UVA 378</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                              |
| The Stern-Brocot N      | <a href="#">UVA 10077</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                              |
| Mr. Kitayuta's Colo     | <a href="#">CF505-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Muntaser Abukadeja</a>                      |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                              |
| DZY Loves Chemis        | <a href="#">CF445-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                              |
| Kolya and Tanya         | <a href="#">CF584-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Yahia Ashraf</a>                            |
| Suffix Structures       | <a href="#">CF448-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Salah</a>                           |
| Complete the Worc       | <a href="#">CF716-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Salah</a>                           |
| Sea and Islands         | <a href="#">CF544-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Salah</a>                           |
| Hopscotch               | <a href="#">CF141-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Yahia Ashraf</a>                            |
| Valera and Contes       | <a href="#">CF369-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Yahia Ashraf</a>                            |
| Bear and Friendshi      | <a href="#">CF791-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Salah</a>                           |
| Preparing Olympia       | <a href="#">CF550-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - SolverToBe (Java)</a>                           |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                              |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Focused and Diffused Thinking</a>                        |
|                         |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Graph Theory - MST - Kruskal</a>                         |
| Highways                | <a href="#">UVA 10147</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mahmoud Adel</a>                            |
| ACM contest and E       | <a href="#">UVA 10600</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Moaz Rashad</a>                             |
| Virtual Friends         | <a href="#">UVA 11503</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Moaz Rashad</a>                             |
| Arctic Network          | <a href="#">UVA 10369</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                              |

|                      | Problem Code                 | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments About your approach                                                                                                                      |
|----------------------|------------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | AC Averages =>               | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                                                                                                                                          |
| Trees on the level   | <a href="#">UVA 122</a>      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - SolverToBe (Java)</a>                                                                                                         |
| Final Standings      | <a href="#">TIMUS 1100</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Stable sort exercise</a>                                                                                                                       |
| Farm                 | <a href="#">TIMUS 1349</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Learn Fermat's Last Theorem</a>                                                                                                                |
|                      |                              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Mashmikh and To      | <a href="#">CF415-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Salma Yehia</a>                                                                                                           |
| Approximating a C    | <a href="#">CF602-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Salma Yehia</a>                                                                                                           |
| Gena's Code          | <a href="#">CF614-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| OR in Matrix         | <a href="#">CF486-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Fox And Two Dots     | <a href="#">CF510-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Adel</a>                                                                                                          |
| Routine Problem      | <a href="#">CF337-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Adel</a>                                                                                                          |
| Vasya and Wrestlin   | <a href="#">CF493-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Hamming Distance     | <a href="#">CF608-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Wet Shark and Bis    | <a href="#">CF621-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mahmoud Mabrok</a>                                                                                                        |
| Kefa and Company     | <a href="#">CF580-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - SolverToBe (Java)</a>                                                                                                         |
| Tavas and SaDDa:     | <a href="#">CF535-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Abanob Ashraf</a>                                                                                                         |
|                      |                              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Minimum Ternary S    | <a href="#">CF1009-D12-B</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                      | <a href="#">CF1030-D12-B</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                      | <a href="#">CF1051-D2-B</a>  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                      | <a href="#">CF1237-D12-B</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                      |                              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                      |                              |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Intro to Greedy</a>                                                                                                                    |
| Painting Eggs        | <a href="#">CF282-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Pasha Maximizes      | <a href="#">CF435-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Hossam Yehia</a>                                                                                                          |
| Little Girl and Gam  | <a href="#">CF276-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Hossam Yehia</a>                                                                                                          |
| Pasha and String     | <a href="#">CF525-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Hossam Yehia</a>                                                                                                          |
| Booking System       | <a href="#">CF416-D2-C</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Vanya and Exams      | <a href="#">CF492-D2-C</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| The Skyline Proble   | <a href="#">UVA 105</a>      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Hanoi Tower Trout    | <a href="#">UVA 10276</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mahmoud Adel</a>                                                                                                          |
| Maze Exploration     | <a href="#">UVA 784</a>      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mahmoud Adel</a>                                                                                                          |
| IP-TV                | <a href="#">UVA 1174</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                      |                              |        |              |                 |                  |                |               | 0             |                   |              |          | Before moving to another sheet, email me with feedback about these problems selection.                                                                     |
|                      |                              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                      |                              |        |              |                 |                  |                |               | 0             |                   |              |          | You don't have to or encouraged to solve the next problem. If you felt you need so, try some of them. Or Proceed to next and solve in parallel, up to you. |
|                      |                              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Devu, the Dumb G     | <a href="#">CF439-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                       |
| Find The Bone        | <a href="#">CF796-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                       |
| Regular Bracket St   | <a href="#">CF26-D12-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                       |
| Inbox (100500)       | <a href="#">CF465-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Different is Good    | <a href="#">CF672-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Permutation          | <a href="#">CF137-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Little Elephant and  | <a href="#">CF259-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Airport              | <a href="#">CF218-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Commen --- The Be    | <a href="#">CF732-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Prison Transfer      | <a href="#">CF427-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| A and B and Comp     | <a href="#">CF519-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Letter               | <a href="#">CF43-D2-B</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Game of Robots       | <a href="#">CF670-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                      |                              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| African Crossword    | <a href="#">CF90-D2-B</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Cows and Poker G     | <a href="#">CF284-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Find Marble          | <a href="#">CF285-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Interesting drink    | <a href="#">CF706-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Megacity             | <a href="#">CF424-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Beautiful Paintings  | <a href="#">CF651-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Ilya and Queries     | <a href="#">CF313-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Code Parsing         | <a href="#">CF255-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Hungry Sequence      | <a href="#">CF327-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Chloe and the seq    | <a href="#">CF743-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Luxurious Houses     | <a href="#">CF581-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                      |                              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Settlers' Training   | <a href="#">CF63-D2-B</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Far Relative's Prot  | <a href="#">CF629-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Wilbur and Array     | <a href="#">CF596-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Text Document An     | <a href="#">CF723-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Shower Line          | <a href="#">CF431-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Misha and Changin    | <a href="#">CF501-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Coat of Anticubism   | <a href="#">CF667-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Ternary Logic        | <a href="#">CF136-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Counting Rhombi      | <a href="#">CF189-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Pashmak and Flow     | <a href="#">CF459-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                      |                              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| The Monster and th   | <a href="#">CF592-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| The Fibonacci Seg    | <a href="#">CF365-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Spider Man           | <a href="#">CF705-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Little Robber Girl's | <a href="#">CF686-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Unary                | <a href="#">CF133-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Canvas Frames        | <a href="#">CF127-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Ohana Cleans Up      | <a href="#">CF554-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Garland              | <a href="#">CF408-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Petya and Staircas   | <a href="#">CF362-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Equidistant String   | <a href="#">CF545-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Vanya and Food P     | <a href="#">CF677-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Calendar             | <a href="#">CF304-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Amr and Pins         | <a href="#">CF507-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Polo the Penguin a   | <a href="#">CF289-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                      |                              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| George and Round     | <a href="#">CF387-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Alyona and flowers   | <a href="#">CF740-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Urbanization         | <a href="#">CF735-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Testing Pants for S  | <a href="#">CF104-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Cells Not Under At   | <a href="#">CF701-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Vanya and Books      | <a href="#">CF552-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Worms                | <a href="#">CF474-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Fortune Telling      | <a href="#">CF59-D2-B</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |

|                         | Problem Code               | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments<br>About your approach |
|-------------------------|----------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|------------------------------------------|
|                         | AC Averages =>             | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                        |
| Walking in the Rain     | <a href="#">CF192-D2-B</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Dima and To-do List     | <a href="#">CF366-D2-B</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Sail                    | <a href="#">CF298-D2-B</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Fox and Cross           | <a href="#">CF389-D2-B</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Rebranding              | <a href="#">CF591-D2-B</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Increase and Decrease   | <a href="#">CF246-D2-B</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Alyona and Mex          | <a href="#">CF682-D2-B</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Coins                   | <a href="#">CF58-D2-B</a>  |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Berland National List   | <a href="#">CF567-D2-B</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Art Union               | <a href="#">CF416-D2-B</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
|                         |                            |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| BerSU Ball              | <a href="#">CF489-D2-B</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Random Teams            | <a href="#">CF478-D2-B</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Friends                 | <a href="#">CF94-D2-B</a>  |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| War of the Corporations | <a href="#">CF625-D2-B</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Road Construction       | <a href="#">CF330-D2-B</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Binary Number           | <a href="#">CF92-D2-B</a>  |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Before an Exam          | <a href="#">CF4-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Running Student         | <a href="#">CF9-D2-B</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Anton and currencies    | <a href="#">CF508-D2-B</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Phone Numbers           | <a href="#">CF151-D2-B</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |

| Problem Name         | Problem Code                      | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments About your approach                                                                                                                                                                       |
|----------------------|-----------------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | AC Averages =>                    | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                                                                                                                                                                                           |
|                      |                                   |        |              |                 |                  |                |               |               |                   |              |          | DON'T Skip colored problems. Don't skip others unless a block is really easy for you                                                                                                                        |
|                      |                                   |        |              |                 |                  |                |               |               |                   |              |          | Remove the given link and write a comment. Start your comment with a classification for the problem: <b>Useless, repeated idea, boring, normal, good problem, interesting problem or important problem.</b> |
| Drazil and Factorial | <a href="#">CF515-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                                                                                                                                            |
| Lucky Permutation    | <a href="#">CF304-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Soldier and Cards    | <a href="#">CF546-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Watchmen             | <a href="#">CF651-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Thinking - Concretely - Symbolically - Pictorially</a>                                                                                                                                  |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Thinking - Problem Constraints</a>                                                                                                                                                      |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Number Theory - Primes</a>                                                                                                                                                              |
| Fox Dividing Chees   | <a href="#">CF371-D2-B</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Abanob Ashraf</a>                                                                                                                                                          |
| Duff in Love         | <a href="#">CF588-D2-B</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Twin Primes          | <a href="#">UVA 10394</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Summation of Four    | <a href="#">UVA 10168 [1]</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Moaz Rashad [2]</a>                                                                                                                                                        |
| The Lottery          | <a href="#">UVA 10325</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Amr Bahaa [3]</a>                                                                                                                                                          |
| Hamburgers           | <a href="#">CF371-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Mint                 | <a href="#">UVA 10717</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                                                                         |
| BITMAP - Bitmap      | <a href="#">SPOJ BITMAP</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
|                      | <a href="#">CODECHEF GCDM</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol uses __int128 to avoid overflow</a>                                                                                                                                                         |
|                      | <a href="#">UVA 10843</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Theory result to read</a>                                                                                                                                                                       |
| The Child and Set    | <a href="#">CF 937-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Tanya and Postcan    | <a href="#">CF 318-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Greg and Array       | <a href="#">CF296-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| The World is a The   | <a href="#">CF131-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Youssef Ali</a>                                                                                                                                                            |
| Trains               | <a href="#">CF88-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                        |
| Semifinals           | <a href="#">CF 318-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Towers               | <a href="#">CF 478-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Algebra - Number Bases and Polynomials</a>                                                                                                                                              |
| To Carry or not to C | <a href="#">UVA 10469</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                                                                         |
| Beat the Spread!     | <a href="#">UVA 10812</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
|                      |                                   |        |              |                 |                  |                |               |               |                   |              |          |                                                                                                                                                                                                             |
| Summation of Poly    | <a href="#">UVA 10302</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Polly the Polynomi   | <a href="#">UVA 498</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
|                      | <a href="#">UVA 11053</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Find O(n) Solution</a>                                                                                                                                                                          |
|                      | <a href="#">LiveArchive 8078</a>  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                                                                         |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Algebra - Patterns in Sequences</a>                                                                                                                                                     |
| R U Kidding Mr. Fe   | <a href="#">UVA 10509</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Wandering Queen      | <a href="#">SPOJ QUEEN</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol to read</a>                                                                                                                                                                                 |
| Spreadsheet          | <a href="#">UVA 196</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
|                      | <a href="#">HACKR sherlock-an</a> |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                                                                         |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Algebra - Summations</a>                                                                                                                                                                |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Algebra - Basic Matrix Operations</a>                                                                                                                                                   |
| Searching for Grap   | <a href="#">CF402-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Flying Saucer Segr   | <a href="#">CF227-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Vasya and Petya's    | <a href="#">CF577-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Round Table Knigh    | <a href="#">CF71-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Kefa and Park        | <a href="#">CF580-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                                                        |
| Knight Tournament    | <a href="#">CF357-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Special Offer! Supe  | <a href="#">CF 318-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Thinking - Problem Abstraction</a>                                                                                                                                                      |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Thinking - Problem Reverse</a>                                                                                                                                                          |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Search Techniques - Backtracking</a>                                                                                                                                                    |
| Graph Coloring       | <a href="#">UVA 193</a>           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                                                                                                                                            |
| 23 out of 5          | <a href="#">UVA 10344</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Nasser</a>                                                                                                                                                         |
| 8 Queens Chess P     | <a href="#">UVA 750</a>           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ayman Salah</a>                                                                                                                                                            |
| Assemble             | <a href="#">UVA 12124</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                                                                         |
|                      | <a href="#">SPOJ FUNPROB</a>      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                                                                         |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Magic Formulas       | <a href="#">CF424-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Pythagorean Triple   | <a href="#">CF707-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Gerald's Hexagon     | <a href="#">CF580-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Points on Line       | <a href="#">CF252-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Find Maximum         | <a href="#">CF353-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Jzzhu and Sequenc    | <a href="#">CF 106-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Simple Game          | <a href="#">CF 318-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Prime Matrix         | <a href="#">CF 371-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Review bitmasking</a>                                                                                                                                                                           |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - DP - Subset Style</a>                                                                                                                                                                   |
| Vacation             | <a href="#">UVA 10192</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | Explained in the tutorial videos                                                                                                                                                                            |
| Dividing coins       | <a href="#">UVA 562</a>           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ayman Salah</a>                                                                                                                                                            |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - DP - Consecutive Ranges Style</a>                                                                                                                                                       |
|                      | <a href="#">SRM149-D1-500</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
|                      | <a href="#">SRM536-D2-1000</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| The Blocks Problem   | <a href="#">UVA 101</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Divisibility         | <a href="#">UVA 10036</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
|                      | <a href="#">UVA 11628</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                                                                         |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Rational Resistanc   | <a href="#">CF344-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| k-Multiple Free Set  | <a href="#">CF275-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Polycarpus' Dice     | <a href="#">CF534-D2-C</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                                                                         |
| Print Check          | <a href="#">CF 937-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Playing Cubes        | <a href="#">CF 207-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| T-primes             | <a href="#">CF 298-D2-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - DP - Nested Ranges Style</a>                                                                                                                                                            |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - DP - General Ranges Style</a>                                                                                                                                                           |
| Creating Palindrom   | <a href="#">UVA 11753</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Aya Elymany</a>                                                                                                                                                            |
| Again Palindrome     | <a href="#">UVA 10617</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol to read</a>                                                                                                                                                                                 |
| Exploring Pyramids   | <a href="#">UVA 1362</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ayman Salah</a>                                                                                                                                                            |
| Cutting Sticks       | <a href="#">UVA 10003</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
| Optimal Array Multi  | <a href="#">UVA 348</a>           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                                                                         |
| Accordion Patience   | <a href="#">UVA 127</a>           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Moaz Rashad</a>                                                                                                                                                            |
| Software CRC         | <a href="#">UVA 128</a>           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Moaz Rashad</a>                                                                                                                                                            |
|                      | <a href="#">ZOJ 1200</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |
|                      |                                   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                                                             |

| Problem Name        | Problem Code                       | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments About your approach                                                                                                                                      |
|---------------------|------------------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | AC Averages =>                     | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                                                                                                                                                          |
| Maze                | <a href="#">CF378-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Thor                | <a href="#">CF705-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Hard problem        | <a href="#">CF706-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Unusual Product     | <a href="#">CF405-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Palindrome Transf   | <a href="#">CF486-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Removing Column     | <a href="#">CF496-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                                                                                                           |
| Crazy Town          | <a href="#">CF499-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                                                                                                           |
| Queue               | <a href="#">CF495-D2-B</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Vika and Squares    | <a href="#">CF510-D2-B</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Cosmic Tables       | <a href="#">CF528-D2-B</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
|                     |                                    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Thinking - Incrementally</a>                                                                                                                           |
|                     |                                    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Thinking - Problem Domain re-interpretation</a>                                                                                                        |
|                     |                                    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Number Theory - Factorization</a>                                                                                                                      |
| Prime Factors       | <a href="#">UVA 583</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Mr. Azad and his S  | <a href="#">UVA 10490</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol to read</a>                                                                                                                                                |
| Perfect P-th Power  | <a href="#">UVA 10622</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Moaz Rashad</a>                                                                                                                           |
| Prime Land          | <a href="#">UVA 516</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
|                     | <a href="#">UVA 10920</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
|                     | <a href="#">SRM274-D1-500</a>      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
|                     |                                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Text Editor         | <a href="#">CF253-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Alternative Thinkin | <a href="#">CF604-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Tennis Champions    | <a href="#">CF735-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Guess Your Way O    | <a href="#">CF507-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                                                                                                           |
| Biathlon            | <a href="#">CF84-D2-C</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Marina and Vasya    | <a href="#">CF584-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Divide by Three     | <a href="#">CF792-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                       |
| Covered Path        | <a href="#">CF308-D2-B</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Facetook Priority W | <a href="#">CF76-D2-B</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Treasure Hunt       | <a href="#">CF303-D2-B</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
|                     |                                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| How Many Points     | <a href="#">UVA 10790</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                                        |
| Factovisors         | <a href="#">UVA 10139</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol to read</a>                                                                                                                                                |
| Fractions Again?!   | <a href="#">UVA 10976</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol to read</a>                                                                                                                                                |
| Cut Ribbon          | <a href="#">CF189-D2-A</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                       |
|                     |                                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
|                     |                                    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Probability - First 9 videos</a>                                                                                                                       |
| Cows and Cars       | <a href="#">UVA 10491</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Revise Probability</a>                                                                                                                                         |
| What is the Probab  | <a href="#">UVA 10056</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                                        |
| Let's Dance         | <a href="#">UVA 10218</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                                        |
| Probability(Given   | <a href="#">UVA 11181</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                                        |
| Another lottery     | <a href="#">UVA 11628</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                                        |
| Airplane            | <a href="#">UVA 12461</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol to read</a>                                                                                                                                                |
|                     | <a href="#">HACKR tower-3-col</a>  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Learn Fermat's little theorem</a>                                                                                                                              |
|                     | <a href="#">CF445-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
|                     | <a href="#">HACKR a-circle-and</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
|                     | <a href="#">UVA 11573</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Learn 0/1 BFS</a>                                                                                                                                              |
| Rankings            | <a href="#">UVA 12263</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                                        |
|                     |                                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Hacker, pack your   | <a href="#">CF822-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                       |
| The Meaningless C   | <a href="#">CF834-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                       |
| Star sky            | <a href="#">CF835-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                       |
|                     |                                    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Before moving to another sheet, email me with feedback about these problems selection.</a>                                                                     |
|                     |                                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
|                     |                                    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">You don't have to or encouraged to solve the next problem. If you felt you need so, try some of them. Or Proceed to next and solve in parallel, up to you.</a> |
|                     |                                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Diverse Permutatio  | <a href="#">CF483-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Replacement         | <a href="#">CF136-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Homework            | <a href="#">CF102-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Little Elephant and | <a href="#">CF221-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Developing Skills   | <a href="#">CF581-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Maxim and Discour   | <a href="#">CF262-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Fox and Box Accur   | <a href="#">CF389-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Ice Skating         | <a href="#">CF218-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Valera and Tubes    | <a href="#">CF441-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Secret              | <a href="#">CF271-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Key Task            | <a href="#">SPOJ CERC07K</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Cleaning Robot      | <a href="#">SPOJ CLEANRBT</a>      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
|                     |                                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| They Are Everywh    | <a href="#">CF701-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Monitor             | <a href="#">CF16-D2-C</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| System Administrat  | <a href="#">CF22-D2-C</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Lucky Sum           | <a href="#">CF122-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| NP-Hard Problem     | <a href="#">CF688-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Vladik and fractio  | <a href="#">CF743-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Case of Matryoshk   | <a href="#">CF556-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Vanya and Label     | <a href="#">CF677-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Exams               | <a href="#">CF479-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Boredom             | <a href="#">CF456-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Learning Language   | <a href="#">CF278-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Beautiful Sets of P | <a href="#">CF268-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
|                     |                                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Strategic Defense   | <a href="#">UVA 497</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Explained in the tutorial videos</a>                                                                                                                           |
| String to Palindrom | <a href="#">UVA 10739</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Explained in the tutorial videos</a>                                                                                                                           |
| Trouble of 13-Dots  | <a href="#">UVA 10819</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Sagheer and Nubia   | <a href="#">CF812-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                                       |
| Purification        | <a href="#">CF330-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Division into Team  | <a href="#">CF149-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Disposition         | <a href="#">CF49-D2-C</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Mashmikh and Nu     | <a href="#">CF415-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Statues             | <a href="#">CF129-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Inna and Huge Car   | <a href="#">CF400-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Anagram Search      | <a href="#">CF144-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Ilya and Sticks     | <a href="#">CF525-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |
| Day at the Beach    | <a href="#">CF599-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                                            |

| Problem Name          | Problem Code               | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments<br>About your approach |
|-----------------------|----------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|------------------------------------------|
|                       | AC Averages =>             | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                        |
|                       |                            |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Appleman and Toa      | <a href="#">CF462-D2-C</a> |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                      |
| Anya and Smartph      | <a href="#">CF518-D2-C</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Little Girl and Maxir | <a href="#">CF276-D2-C</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Sereja and Algori     | <a href="#">CF368-D2-C</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| The Child and Toy     | <a href="#">CF437-D2-C</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Perfect Pair          | <a href="#">CF318-D2-C</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Another Problem o     | <a href="#">CF165-D2-C</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Socks                 | <a href="#">CF731-D2-C</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Valera and Election   | <a href="#">CF369-D2-C</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |

| Problem Name                  | Problem Code                   | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments<br>About your approach                                         |
|-------------------------------|--------------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|----------------------------------------------------------------------------------|
|                               | AC Averages =>                 | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                                                                |
|                               |                                |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Thinking - Search Space and Output Analysis</a>              |
|                               |                                |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Thinking - Observations Discovery</a>                        |
|                               |                                |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Game Theory - Intro</a>                                      |
| Win or Freeze                 | <a href="#">CF151-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                 |
| Euclid's Game                 | <a href="#">UVA 10368</a>      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Moaz Rashad</a>                                 |
| Pyramids                      | <a href="#">SPOJ PIR</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                              |
| Power of Cryptograph          | <a href="#">UVA 113</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol to read</a>                                                      |
|                               | <a href="#">SRM458-D2-500</a>  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Is There A Second Value       | <a href="#">UVA 10462</a>      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               | <a href="#">SRM381-D2-1000</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               |                                |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Modified GCD                  | <a href="#">CF75-D2-C</a>      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                 |
| Alyona and mex                | <a href="#">CF740-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                 |
| Hamburgers                    | <a href="#">CF371-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Wet Shark and Flow            | <a href="#">CF621-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Predict Outcome of Toss       | <a href="#">CF451-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Balls and Boxes               | <a href="#">CF260-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                 |
| Alice and Bob                 | <a href="#">CF347-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Nasser</a>                              |
| Mahmoud and Ehab              | <a href="#">CF959-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Salah</a>                               |
| Almost Equal                  | <a href="#">CF1206-D2-C</a>    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                 |
|                               | <a href="#">CF1220-D12-C</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               | <a href="#">CF1065-D2-C</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               | <a href="#">CF1036-D2-C</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               | <a href="#">CF1068-D2-C</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               | <a href="#">CF313-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Balls Game                    | <a href="#">CF430-D2-B</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Magical Array                 | <a href="#">CF266-D2-B</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               |                                |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Thinking - Misc - Solution Verification - Implementation</a> |
|                               |                                |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Graph Theory - Dijkstra</a>                                  |
| Jugs                          | <a href="#">UVA 571</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                 |
| Sending email                 | <a href="#">UVA 10986</a>      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Lift Hopping                  | <a href="#">UVA 10801</a>      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Shopping                      | <a href="#">SPOJ SHOP</a>      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Ordering                      | <a href="#">UVA 872</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               | <a href="#">CF1084-D2-C</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               | <a href="#">CF1059-D2-C</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               | <a href="#">CF101933-GYM-K</a> |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                              |
| Karen and Coffee              | <a href="#">CF816-D2-B</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                 |
|                               |                                |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Bulls and Cows                | <a href="#">CF63-D2-C</a>      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                              |
| Xor-tree                      | <a href="#">CF430-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Median Smoothing              | <a href="#">CF591-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Coloring Trees                | <a href="#">CF711-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be</a>                                    |
| Clear Symmetry                | <a href="#">CF202-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               | <a href="#">CF1237-D12-C2</a>  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Sereja and Mirroring          | <a href="#">CF426-D2-B</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Restoring Painting            | <a href="#">CF275-D2-B</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               |                                |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               |                                |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Computational Geometry - Lines Intersections</a>             |
| Gleaming the Cubes            | <a href="#">UVA 737</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                              |
| Intersecting Line Segments    | <a href="#">UVA 866</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                              |
|                               |                                |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Computational Geometry - Circles</a>                         |
| The Circumference of a Circle | <a href="#">UVA 438</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                              |
| Points in Figures: Rectangles | <a href="#">UVA 477</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                              |
| Square Pegs And Rectangles    | <a href="#">UVA 356</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol to read</a>                                                      |
|                               | <a href="#">UVA 453</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Learn Handling Precisions</a>                                        |
| Divisibility of Factors       | <a href="#">UVA 10484</a>      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol to read</a>                                                      |
|                               | <a href="#">SRM436-D2-500</a>  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               | <a href="#">CF975-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               | <a href="#">CF1047-D2-C</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               | <a href="#">CF1075-D2-C</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               | <a href="#">CF758-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               | <a href="#">UVA 10525</a>      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Sol. Also solvable in 2 other ways.</a>                        |
|                               |                                |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Prime Permutation             | <a href="#">CF124-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Hometask                      | <a href="#">CF155-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Terse princess                | <a href="#">CF148-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Nasser</a>                              |
| Hacking Cypher                | <a href="#">CF490-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Dreamoon and Sum of Digits    | <a href="#">CF476-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                 |
| Try and Catch                 | <a href="#">CF195-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Editorial - Eng Ahmed Osama</a>                                      |
| Primes or Palindromes         | <a href="#">CF569-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| View Angle                    | <a href="#">CF257-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Editorial - Eng Ahmed Osama</a>                                      |
| Little Pony and Sort by Shift | <a href="#">CF460-D2-B</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               |                                |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Thinking - Error Inspection - History - Contest Strategy</a> |
|                               |                                |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - DP - Building Output</a>                                     |
| Unidirectional TSP            | <a href="#">UVA 116</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Make Palindrome               | <a href="#">UVA 10453</a>      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                              |
| Fast Food                     | <a href="#">UVA 662</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Palindromic Subsequences      | <a href="#">UVA 11404</a>      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Gone Fishing                  | <a href="#">UVA 757</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol to read</a>                                                      |
| Special Olympics              | <a href="#">CF199-D2-B</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Rings and Glue                | <a href="#">UVA 10301</a>      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                              |
|                               |                                |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - DP - Counting</a>                                            |
| k-Tree                        | <a href="#">CF431-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                             |
| Caesar's Legions              | <a href="#">CF118-D2-D</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| UnsealTheSafe                 | <a href="#">SRM354-D2-1000</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| DiceGames                     | <a href="#">SRM349-D1-500</a>  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
|                               | <a href="#">SPOJ TWINSNOW</a>  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol - text clarification</a>                                         |
|                               | <a href="#">SPOJ FACENEMY</a>  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                              |
|                               |                                |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| No to Palindromes!            | <a href="#">CF465-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Triangle                      | <a href="#">CF408-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| To Add or Not to Add          | <a href="#">CF231-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Number of Ways                | <a href="#">CF466-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                             |
| Queue                         | <a href="#">CF141-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |
| Magical Boxes                 | <a href="#">CF270-D2-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                  |



| Problem Name                 | Problem Code                       | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments<br>About your approach                                                                                                                   |
|------------------------------|------------------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | AC Averages =>                     | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                                                                                                                                          |
| Find Pair                    | <a href="#">CF160-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Multitasking                 | <a href="#">CF388-D2-B</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Non-square Equation          | <a href="#">CF333-D2-B</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                              |                                    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Thinking - Let's Put All Together</a>                                                                                                  |
|                              |                                    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - DP - Table Method</a>                                                                                                                  |
|                              |                                    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Graph Theory - Floyd Warshal</a>                                                                                                       |
| Frogger                      | <a href="#">UVA 534</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                        |
| Identifying Concurrency      | <a href="#">UVA 334</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Numbering Paths              | <a href="#">UVA 125</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                        |
| Jack Straws                  | <a href="#">UVA 273</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                        |
| Longest Match                | <a href="#">UVA 10100</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Isolated Segments            | <a href="#">UVA 11343</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                        |
| Counting                     | <a href="#">UVA 10198</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Needs Big Integer: Have it in your c++ library or learn Java for these (rare) cases</a>                                                        |
|                              |                                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Mafia                        | <a href="#">CF349-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Sereja and Prefixes          | <a href="#">CF381-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| About Bacteria               | <a href="#">CF199-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| DNA Alignment                | <a href="#">CF520-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Geometric Progression        | <a href="#">CF567-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Watering Flowers             | <a href="#">CF617-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Quiz                         | <a href="#">CF337-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Secret Combination           | <a href="#">CF706-D2-B</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| MUH and Important            | <a href="#">CF401-D2-B</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Lucky Mask                   | <a href="#">CF746-D2-B</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                              |                                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                              |                                    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Measuring Algorithms Performance - 2</a>                                                                                               |
|                              |                                    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Graph Theory - Tree Diameter and Isomorphism</a>                                                                                       |
| PT07Z                        | <a href="#">SPOJ PT07Z</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                        |
| Roads in the North           | <a href="#">UVA 10308</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                        |
| Subway tree system           | <a href="#">LIVEARCHIVE 2935</a>   |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                        |
|                              |                                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Shaass and Lights            | <a href="#">CF294-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                                                                                           |
| Journey                      | <a href="#">CF721-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Captain Marmot               | <a href="#">CF474-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>                                                                                                           |
| The Big Race                 | <a href="#">CF592-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Molly's Chemicals            | <a href="#">CF778-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>                                                                                                       |
| Anatoly and Cockroaches      | <a href="#">CF718-D2-B</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Opposites Attract            | <a href="#">CF131-D2-B</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                              |                                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Railway                      | <a href="#">UVA 10263</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol to read</a>                                                                                                                                |
| Factorial Factors            | <a href="#">UVA 884</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Wifi Access                  | <a href="#">UVA 12748</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                        |
| Lining Up                    | <a href="#">UVA 270</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Nasser. Don't Code O(N^3)</a>                                                                                     |
| Pouring water                | <a href="#">SPOJ POUR1</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Moaz Rashad</a>                                                                                                           |
|                              | <a href="#">CF23-D12-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                              | <a href="#">CF869-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                              | <a href="#">SRM321-D1-500</a>      |        |              |                 |                  |                |               | 0             |                   |              |          | See Rushiose's code in arena summary                                                                                                                       |
|                              |                                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                              |                                    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch Video - Expected Value</a>                                                                                                               |
| God, Save me                 | <a href="#">UVA 10777</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                        |
|                              | <a href="#">CF839-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                              | <a href="#">CF454-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                              | <a href="#">SRM577-D1-250</a>      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Editorial</a>                                                                                                                                  |
|                              | <a href="#">HACKR lazy-sorting</a> |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Revise Expected Value</a>                                                                                                                      |
|                              | <a href="#">SPOJ ALIENS</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol - Practice on min enclosing circle</a>                                                                                                     |
|                              | <a href="#">CF340-D2-B</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                              |                                    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Before moving to another sheet, email me with feedback about these problems selection.</a>                                                     |
|                              |                                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                              |                                    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Optional Problems</a>                                                                                                                          |
|                              |                                    |        |              |                 |                  |                |               | 0             |                   |              |          | You don't have to or encouraged to solve the next problem. If you felt you need so, try some of them. Or Proceed to next and solve in parallel, up to you. |
|                              |                                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Checkposts                   | <a href="#">CF427-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Literature Lesson            | <a href="#">CF139-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Arpa's loud Owf and          | <a href="#">CF742-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Parity Game                  | <a href="#">CF298-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Beauty Pageant               | <a href="#">CF246-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Heroes                       | <a href="#">CF80-D2-C</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Dynasty Puzzles              | <a href="#">CF192-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Buns                         | <a href="#">CF106-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Counting Kangaroos           | <a href="#">CF373-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Corporation Mail             | <a href="#">CF56-D2-C</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Matrix                       | <a href="#">CF365-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Pick up sticks               | <a href="#">UVA 11686</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                                                                                        |
|                              |                                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Little Elephant and Integers | <a href="#">CF205-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Sereja and Contest           | <a href="#">CF315-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Vasya and Robot              | <a href="#">CF355-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Hockey                       | <a href="#">CF86-D2-C</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Petya and File System        | <a href="#">CF66-D2-C</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Kyoya and Colored Flowers    | <a href="#">CF554-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| George and Job               | <a href="#">CF467-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Harmony Analysis             | <a href="#">CF610-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Anton and Making Pairs       | <a href="#">CF734-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Table Decorations            | <a href="#">CF478-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Recycling Bottles            | <a href="#">CF672-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
|                              |                                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Message                      | <a href="#">CF157-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Wilbur and Points            | <a href="#">CF596-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Cows and Sequence            | <a href="#">CF284-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Ladder                       | <a href="#">CF279-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Not Wool Sequence            | <a href="#">CF239-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Anagram                      | <a href="#">CF254-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| DZY Loves Sequences          | <a href="#">CF447-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| DZY Loves Physics            | <a href="#">CF445-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Misha and Forest             | <a href="#">CF501-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |
| Jzzhu and Chocolate          | <a href="#">CF450-D2-C</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                                                                                            |

| Problem Name       | Problem Code               | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments<br>About your approach |
|--------------------|----------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|------------------------------------------|
|                    | AC Averages =>             | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                        |
| Cinema             | <a href="#">CF670-D2-C</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
|                    |                            |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Report             | <a href="#">CF631-D2-C</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Bear and Prime Num | <a href="#">CF385-D2-C</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Robbery            | <a href="#">CF90-D2-C</a>  |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Vasya and Basketba | <a href="#">CF493-D2-C</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Vanya and Scales   | <a href="#">CF552-D2-C</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Pashmak and Buses  | <a href="#">CF459-D2-C</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |
| Fancy Number       | <a href="#">CF118-D2-C</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                          |

| Problem Name         | Problem Code     | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments About your approach                          |
|----------------------|------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|----------------------------------------------------------------|
|                      | AC Averages ==>  | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                                              |
| Dividing Island      | CF63-D2-D        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Flowers              | CF474-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>           |
| Dima and Bacteria    | CF400-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | CF1043-D12-C     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | CF1033-D12-C     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | CF1066-D3-E      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | CF534-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | CF899-D2-E       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | CF729-D12-D      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Tourist Problem      | CF398-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Lorenzo Von Matter   | CF381-D4-E       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Restore Graph        | CF349-D3-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | CF309-D1-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | CF101-D1-B       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |
|                      | SRM569-D2-1000   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | CF961-D12-D      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | CF955-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | UVA 12869        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |
|                      | CF372-D1-B       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      |                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Data Structures - Segment Tree (2 vid)</a> |
| Interval Product     | UVA 12532        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Potentiometers       | LIVEARCHIVE 2191 |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Halt The War         | SPOJ CDC12_H     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Counting Primes      | SPOJ CNTPRIME    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Horrible Queries     | SPOJ HORRIBLE    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Light Switching      | SPOJ LITE        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Circular RMQ         | CF52-D12-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| A Famous City        | SPOJ CITY2       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |
| RMQ with Shifts      | UVA 12299        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">See sscanf and sprintf usage</a>                   |
| R2D2 and Droid Army  | CF514-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Use rmq</a>                                        |
| Ahoy, Pirates!       | UVA 11402        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |
| Brackets             | SPOJ BRCKTS      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |
| Present              | CF460-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| MessageMess          | SRM149-D1-500    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| DiceGames            | SRM349-D1-500    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Mirror, Mirror       | UVA 466          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Maximum Sum          | SPOJ KGSS        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | SRM297-D1-500    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | SRM441-D1-250    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | CF201-D1-B       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | CF380-D1-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | CF161-D12-D      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Reading: DP on Trees</a>                           |
|                      | CF61-D2-E        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | SPOJ KOMPICI     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      |                  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Quantity of Strings  | CF151-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Eternal Victory      | CF61-D2-D        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Array Division       | CF808-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>           |
|                      | CF45-D12-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | SRM428-D2-1000   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | SGU 321          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |
|                      | CODECHEF OPPOSIT |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | SRM513-D2-1000   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | SRM292-D1-500    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | SRM405-D2-1000   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Hiring Staff         | CF218-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Tavas and Karafs     | CF339-D4-E       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Permutations         | CF186-D3-C       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |
|                      |                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Watch - Two pointers technique</a>                 |
| Spider's Web         | CF216-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Chips                | CF334-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Vasya and String     | CF676-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| The SetStack Comp    | LiveArchive 3634 |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |
| Database             | UVA 1592         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Can you answer the   | SPOJ GSS1        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |
|                      | SPOJ BILLIARD    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |
| Can you answer the   | SPOJ GSS3        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | SPOJ ABA12E      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |
|                      | UVA 11825        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |
|                      | CF472-D12-D      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | UVA 12325        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Prove your Solution</a>                            |
|                      | UVA 12047        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |
|                      | UVA 10705        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |
|                      | UVA 1555         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |
|                      | CF80-D2-D        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      |                  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Mahmoud and a Dict   | CF766-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>           |
| An overnight dance   | CF814-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>           |
| Polyline             | CF617-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Queue                | CF92-D2-D        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | CF1038-D2-D      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | CF552-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | CF101917-D12-E   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | CF1058-D2-D      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | CF1042-D12-D     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
|                      | SPOJ BIA         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |
| Plant                | CF198-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Reberland Linguistic | CF381-D4-E       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Lucky Permutation    | CF287-D3-C       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">DP - Probability</a>                               |
|                      |                  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Dice Throwing        | UVA 10759        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |
| TestBettingStrategy  | SRM339-D1-500    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                |
| Collecting Bugs      | PKU 2096         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |
| France '98           | UVA 542          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                            |

| Problem Name          | Problem Code                     | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments About your approach                  |
|-----------------------|----------------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|--------------------------------------------------------|
|                       | AC Averages ==>                  | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                                      |
| Tribbles              | <a href="#">UVA 11021</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    |
| Tennis contest        | <a href="#">UVA 12457</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    |
| Water Falls           | <a href="#">UVA 833</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Number Sequence       | <a href="#">UVA 10706</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Is It A Tree?         | <a href="#">UVA 615</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Help R2-D2!           | <a href="#">SPOJ.HELPR2D2</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
|                       | <a href="#">CF1016-D2-E</a>      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
|                       | <a href="#">UVA 11997</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    |
|                       | <a href="#">FbHkrCup 18-R1-A</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
|                       | <a href="#">SRM456-D2-1000</a>   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
|                       |                                  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Andrey and Problem    | <a href="#">CF443-D2-D</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    |
| Three Logos           | <a href="#">CF581-D2-D</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Good Sequences        | <a href="#">CF265-D2-D</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Party                 | <a href="#">CF135-D2-E</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Cupboard and Balloons | <a href="#">CF346-D2-E</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Cycles                | <a href="#">CF236-D2-E</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
|                       |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">DP - Masks (2 vid)</a>                     |
| Pebble Solitaire      | <a href="#">UVA 10651</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Kefa and Dishes       | <a href="#">CF580-D2-D</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be</a>          |
| Permutations          | <a href="#">SPOJ.PERMUT1</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Assignments           | <a href="#">SPOJ.ASSIGN</a>      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
|                       | <a href="#">CF16-D2-E</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Count the Faces.      | <a href="#">UVA 10178</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Read first Euler Formula</a>               |
| LCM Cardinality       | <a href="#">UVA 10892</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Robot Rapping Res     | <a href="#">CF645-D12-D</a>      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Wavio Sequence        | <a href="#">UVA 10534</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    |
|                       | <a href="#">UVA 10342</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol - read the statement clarification</a> |
|                       |                                  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Directed Roads        | <a href="#">CF711-D2-D</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Block Tower           | <a href="#">CF327-D2-D</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| A and B and Interes   | <a href="#">CF519-D2-D</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| As Fast As Possible   | <a href="#">CF701-D2-D</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Chloe and pleasant    | <a href="#">CF743-D2-D</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Roads in Berland      | <a href="#">CF265-D2-E</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Photographer          | <a href="#">CF266-D2-E</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| LCM Challenge         | <a href="#">CF236-D2-E</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
|                       |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">String Processing - Trie</a>               |
| Search in the diction | <a href="#">SPOJ.DICT</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Disk Tree             | <a href="#">UVA 1558</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Phone List            | <a href="#">SPOJ.PHONELST</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Cellphone Typing      | <a href="#">UVA 12526</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Vasily's Multiset     | <a href="#">CF706-D2-D</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Exchange Rates        | <a href="#">UVA 10113</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Equation              | <a href="#">UVA 727</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Safe                  | <a href="#">CF47-D2-D</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Central Post Office   | <a href="#">UVA 12379</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    |
| Permalex              | <a href="#">UVA 153</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    |
|                       |                                  |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">DP - Sub-rectangle style</a>               |
|                       | <a href="#">UVA 507</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
|                       | <a href="#">UVA 10667</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
|                       |                                  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Volleyball            | <a href="#">CF96-D2-D</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Lazy Student          | <a href="#">CF606-D2-D</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Multiplication Table  | <a href="#">CF448-D2-D</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solve to be (Java)</a>    |
|                       | <a href="#">CF486-D2-D</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
|                       | <a href="#">CF1040-D2-D</a>      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
|                       | <a href="#">CF264-D1-C</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
|                       | <a href="#">CF506-D1-A</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
|                       | <a href="#">CODECHEF.KSUM</a>    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
|                       | <a href="#">CF623-D1-B</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Divisible by Seven    | <a href="#">CF235-D2-E</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Devu and Partitionin  | <a href="#">CF506-D2-E</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |
| Arthur and Table      | <a href="#">CF293-D2-E</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        |

| Problem Name                  | Problem Code                          | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments<br>About your approach            |
|-------------------------------|---------------------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|-----------------------------------------------------|
|                               | AC Averages =>                        | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                                   |
|                               |                                       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">String Processing - KMP (2 vid)</a>     |
| Oulipo                        | <a href="#">PKU 3481</a>              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| A Needle in the Haystack      | <a href="#">SPOJ NHAY</a>             |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Finding the Tesseract         | <a href="#">SPOJ TESSER</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Period                        | <a href="#">SPOJ PERIOD</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Prefixes and Suffixes         | <a href="#">CF432-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Tavas and Malekas             | <a href="#">CF535-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Be Efficient                  | <a href="#">UVA 11155</a>             |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Vertex Cover                  | <a href="#">SPOJ PT07X</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| First Digit Law               | <a href="#">CF54-D12-C</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">CF500-D12-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">HACKR vertical-sticks</a> |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">UVA 10174</a>             |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">UVA 1333</a>              |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol - Text/Background Clarification</a> |
|                               | <a href="#">CF842-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">CF709-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">SPOJ MSKYCODE</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
|                               | <a href="#">LiveArchive 8015</a>      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
|                               |                                       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Robin Hood                    | <a href="#">CF672-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| End of Exams                  | <a href="#">CF94-D2-D</a>             |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Equivalent Strings            | <a href="#">CF560-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol to learn</a>                        |
| Count Good Substrings         | <a href="#">CF451-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Mushroom Scientists           | <a href="#">CF188-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Analyzing Polyline            | <a href="#">CF195-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">CF1023-D12-E</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">CF1060-D12-C</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Bear and Prime 10             | <a href="#">CF986-D2-C</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Team                          | <a href="#">CF481-D2-C</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               |                                       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">DP - Games (2 vid)</a>                  |
| Bachet's Game                 | <a href="#">UVA 10404</a>             |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| <a href="#">EllysCheckers</a> | <a href="#">SRM534-D1-250</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| <a href="#">RowAndCoins</a>   | <a href="#">SRM522-D1-250</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| <a href="#">BagsOfGold</a>    | <a href="#">SRM228-D1-500</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Bag of mice                   | <a href="#">CF148-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">CF1147-D1-B</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| MELE3                         | <a href="#">SPOJ MELE3</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| Roads                         | <a href="#">SPOJ ROADS</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| The Tree Root                 | <a href="#">UVA 10459</a>             |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| SKYLINE                       | <a href="#">UVA 1232</a>              |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| Ordering the Soldiers         | <a href="#">SPOJ ORDERS</a>           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| Playlist                      | <a href="#">CF268-D2-E</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
|                               | <a href="#">SRM481-D1-500</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               |                                       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Little Girl and Maximum       | <a href="#">CF278-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">See editorials</a>                      |
| Two Strings                   | <a href="#">CF224-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| Big Maximum Sum               | <a href="#">CF75-D2-D</a>             |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">SPOJ BROKTS2</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
|                               | <a href="#">CF1057-D12-C</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">CF1066-D3-F</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">CF1064-D2-E</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">CF459-D2-E</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">UVA 10888</a>             |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">CF1043-D12-D</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Football Championship         | <a href="#">CF255-D2-C</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Given Length and Sum          | <a href="#">CF489-D2-C</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               |                                       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Trip Routing                  | <a href="#">UVA 186</a>               |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| Scheduling Lecture            | <a href="#">UVA 607</a>               |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| Weird Function                | <a href="#">SPOJ WEIRDFN</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| The ? 1 ? 2 ? ... ? 1         | <a href="#">UVA 10025</a>             |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Dictionary Subsequence        | <a href="#">SPOJ DICTSUB</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| Jimmi's Riddles               | <a href="#">UVA 10058</a>             |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| Friends and Subsequences      | <a href="#">CF689-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Sum of Squares w/ Primes      | <a href="#">SPOJ SEGSQLRSS</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| Travel in Desert              | <a href="#">UVA 10816</a>             |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| Almost Union-Find             | <a href="#">UVA 11987</a>             |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
|                               | <a href="#">SRM537-D2-1000</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">CF513-D12-C</a>           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
|                               | <a href="#">SRM453.5-D2-1000</a>      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">SPOJ PARSUMS</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
|                               | <a href="#">CF1138-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               |                                       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Cow Program                   | <a href="#">CF284-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Random Task                   | <a href="#">CF431-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Greg and Graph                | <a href="#">CF296-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Russian Roulette              | <a href="#">CF104-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Bicycle Race                  | <a href="#">CF659-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Greenhouse Effect             | <a href="#">CF270-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">CF645-D12-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">CF459-D2-C</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">CODECHEF REDCGA</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               | <a href="#">CF1005-D3-F</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Pocket Book                   | <a href="#">CF158-D2-C</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Levko and Array Ranges        | <a href="#">CF361-D2-C</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Ice Cave                      | <a href="#">CF256-D2-C</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
|                               |                                       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Robbery                       | <a href="#">UVA 707</a>               |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| The Errant Physicist          | <a href="#">UVA 126</a>               |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| Brackets sequence             | <a href="#">UVA 1626</a>              |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| Unique World                  | <a href="#">UVA 10448</a>             |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>    |
| Bad Luck Island               | <a href="#">CF540-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |
| Shopping Trip                 | <a href="#">UVA 11284</a>             |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
| Hotel booking                 | <a href="#">UVA 11635</a>             |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
|                               | <a href="#">CF337-D2-D</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                 |
|                               | <a href="#">HACKR ajourney</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                     |

| Problem Name                      | Problem Code                         | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments<br>About your approach                 |
|-----------------------------------|--------------------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|----------------------------------------------------------|
|                                   | AC Averages =>                       | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                                        |
|                                   | <a href="#">CF665-D12-E</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
|                                   |                                      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
| Hit Ball                          | <a href="#">CF203-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
| Sereja ans Anagra                 | <a href="#">CF368-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
| Choosing Capital f                | <a href="#">CF219-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
| Coloring Brackets                 | <a href="#">CF149-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
| Cycle in Graph                    | <a href="#">CF263-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
|                                   | <a href="#">CF101187-GYM-F</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | SRM319-D1-500                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
|                                   | <a href="#">AtCoder092-ARC-B</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
|                                   | <a href="#">AtCoder002-AGC-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
| Fixing Typos                      | <a href="#">CF363-D1-C</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
| Cutting Figure                    | <a href="#">CF196-D1-C</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
| Escape from Stone                 | <a href="#">CF265-D1-C</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
|                                   |                                      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
|                                   |                                      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Geometry - Simple and Convex Polygons</a>    |
|                                   |                                      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Geometry - Polygon Area - Centroid - Cut</a> |
| <a href="#">BestTriangulation</a> | SRM278-D2-500                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
| Trees on My Island                | <a href="#">UVA 10088</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
| Packing polygons                  | <a href="#">UVA 10005</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | LIVEARCHIVE 2831                     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Use polygon cut</a>                          |
| Video Surveillance                | <a href="#">UVA 588</a>              |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Use polygon cut</a>                          |
|                                   | SRM514-D1-500                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
|                                   | SRM473-D1-500                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
|                                   | SRM555-D2-1000                       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
|                                   | UVA 557                              |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | SRM285-D1-500                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
| Xrange's Pancake                  | <a href="#">HACKR xrange-and-piz</a> |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | SRM525-D1-500                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
|                                   | UVA 11648                            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | <a href="#">CF101864-GYM-A</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | <a href="#">CF101864-GYM-L</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | <a href="#">CF28-D12-C</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
|                                   |                                      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Geometry - Point in polygon</a>              |
|                                   | UVA 881                              |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | UVA 11665                            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | TIMUS 1599                           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
| Polygons                          | <a href="#">UVA 137</a>              |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   |                                      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Graph Theory - Maximum Flow (2 vid)</a>      |
| Potholes                          | <a href="#">SPOJ POTHOLE</a>         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
| Power Transmissi                  | <a href="#">UVA 10330</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
| Gopher II                         | <a href="#">UVA 10080</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
| Software Allocator                | <a href="#">UVA 259</a>              |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | <a href="#">UVA 10349</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol - 2 ways</a>                             |
|                                   | UVA 12168                            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
| A Plug for UNIX                   | <a href="#">UVA 753</a>              |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | <a href="#">UVA 10349</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol - 2 ways</a>                             |
| Intergalactic Map                 | <a href="#">SPOJ IM</a>              |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | <a href="#">UVA 11159</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | UVA 1194                             |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
| Fence Obstacle C                  | <a href="#">PKU 2374</a>             |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
| River Crossing                    | UVA 10514                            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | SRM368-D1-500                        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | SRM373-D2-1000                       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | SRM558-D1-250                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
|                                   | ZOJ 2587                             |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | SRM550-D2-1000                       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
|                                   | UVA 10180                            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | TIMUS 1156                           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                          |
|                                   | UVA 1184                             |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |
|                                   | UVA 670                              |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                      |

| Problem Name         | Problem Code                         | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments<br>About your approach   |
|----------------------|--------------------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|--------------------------------------------|
|                      | AC Averages =>                       | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                          |
|                      |                                      |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Graph Theory - SCC (2 vid)</a> |
| The Bottom of a G    | <a href="#">SPOJ BOTTOM</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
| Test                 | <a href="#">UVA 10731</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">SRM312-D1-500</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">CF467-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Theseus and labyr    | <a href="#">CF676-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Cunning Gena         | <a href="#">CF418-D1-B</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Sabotage             | <a href="#">UVA 10480</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">SRM352-D2-1000</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Garland              | <a href="#">UVA 1555</a>             |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">CF101589-GYM-F</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">CF1016-D12-D</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">CF26-D12-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol - must read</a>            |
|                      | <a href="#">CF1012-D1-B</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">CF1010-D1-C</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">CF833-D12-D</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">HACKR house-location</a> |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">CF621-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">CF101992-GYM-D</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">SRM608-D2-1000</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      |                                      |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Gifts by the List    | <a href="#">CF681-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| DZY Loves Modifi     | <a href="#">CF447-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          | Prove                                      |
| Mike and Feet        | <a href="#">CF548-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Special Grid         | <a href="#">CF435-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Roman and Numb       | <a href="#">CF401-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Persistent Bookca    | <a href="#">CF707-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
| Regular Bridge       | <a href="#">CF550-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">CF1059-D2-D</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Almost Arithmetica   | <a href="#">CF 355-D2-C</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Title                | <a href="#">CF 386-D2-C</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Treasure             | <a href="#">CF 390-D2-C</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      |                                      |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Unique Attack        | <a href="#">ZOJ 2587</a>             |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| End of Fun           | <a href="#">SPOJ DCEPC12E</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Grammar Evaluati     | <a href="#">UVA 622</a>              |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
| Find the Winning A   | <a href="#">UVA 10111</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
| Check the difficulty | <a href="#">PKU 2151</a>             |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
| Proving Equivalen    | <a href="#">UVA 12167</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
| DDF                  | <a href="#">UVA 547</a>              |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Dominos              | <a href="#">UVA 11504</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">SRM419-D2-1000</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Winning Streak       | <a href="#">UVA 11176</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">SRM391-D2-1000</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">SRM465-D1-500</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">UVA 10740</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">UVA 12261</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">LIVEARCHIVE 4008</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">UVA 1342</a>             |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">CF811-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">AtCoder026-AGC-B</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">SPOJ FISHES</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">UVA 11475</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      |                                      |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Red-Green Tower      | <a href="#">CF478-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Renting Bikes        | <a href="#">CF363-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Lucky Number 2       | <a href="#">CF146-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Tennis Game          | <a href="#">CF496-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Bubble Sort Graph    | <a href="#">CF340-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Upgrading Array      | <a href="#">CF402-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">ZOJ 3305</a>             |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">CF1017-D12-D</a>         |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Game                 | <a href="#">CF 385-D2-C</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Ciel and Robot       | <a href="#">CF 328-D2-C</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Plus and Square R    | <a href="#">CF 708-D2-C</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      |                                      |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Boxes in a Line      | <a href="#">UVA 12657</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">SPOJ QUEST4</a>          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
| Multifactorials      | <a href="#">UVA 11347</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Crimewave            | <a href="#">UVA 563</a>              |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">SRM545-D2-1000</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">SRM495-D1-500</a>        |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Primitive Root       | <a href="#">SPOJ PROOT</a>           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
| Of Zorcs and Axes    | <a href="#">CF101149-GYM-G</a>       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
| Connected Comp       | <a href="#">CF292-D12-D</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| AND Rounds           | <a href="#">SPOJ ANDROUND</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
| Campus Roads         | <a href="#">UVA 11473</a>            |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
| The Child and Zoo    | <a href="#">CF437-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">CF403-D1-C</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">CF787-D2-C</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">CF309-D12-B</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">SRM392-D1-1000</a>       |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">UVA 12128</a>            |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">Timus 1362</a>           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">CF1012-D1-C</a>          |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">SPOJ COCONUTS</a>        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      | <a href="#">FbHkrCup 18-RQ-C</a>     |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
|                      | <a href="#">LIVEARCHIVE 4682</a>     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                        |
|                      |                                      |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Image Preview        | <a href="#">CF651-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Maximum Xor Sec      | <a href="#">CF281-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Ilya and Roads       | <a href="#">CF313-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Mr. Bender and Sc    | <a href="#">CF255-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Fish Weight          | <a href="#">CF298-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| T-decomposition      | <a href="#">CF237-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |
| Wizards and Huge     | <a href="#">CF168-D2-D</a>           |        |              |                 |                  |                |               | 0             |                   |              |          |                                            |

| Problem Name                       | Problem Code                                      | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments About your approach                                                |
|------------------------------------|---------------------------------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|--------------------------------------------------------------------------------------|
|                                    | AC Averages =><br><a href="#">CODECHEF BJUDGE</a> | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                                                                    |
| Dima and Salad                     | <a href="#">CF 369-D2-C</a>                       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    |                                                   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| Arbitrage                          | <a href="#">UVA 104</a>                           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| Random Task                        | <a href="#">CF431-D2-D</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| Black Box                          | <a href="#">UVA 501</a>                           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol - Must Read</a>                                                      |
| Expressions                        | <a href="#">UVA 11234</a>                         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| Showstopper                        | <a href="#">SPOJ MSE07E</a>                       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Read SPOJ users' comments about IO. See here sol</a>                     |
| Tobo or not Tobo                   | <a href="#">SPOJ ANARC08A</a>                     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| Sum-up the Prime:                  | <a href="#">UVA 10419</a>                         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| Largest Rectangle                  | <a href="#">SPOJ HISTOGRAM</a>                    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol. Don't implement as adhoc/greedy/Pure STL. Use a data structure.</a> |
|                                    | <a href="#">UVA 663</a>                           |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| <a href="#">KingdomReorganize</a>  | <a href="#">SRM531-D2-1000</a>                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| The Problem with I                 | <a href="#">UVA 10092</a>                         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| Psycho                             | <a href="#">SPOJ.PSYCHON</a>                      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| Minimal Ratio Tree                 | <a href="#">LIVEARCHIVE 4326</a>                  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| RACING                             | <a href="#">UVA 1234</a>                          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| <a href="#">ActivateGame</a>       | <a href="#">SRM470-D2-1000</a>                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| Pair of Numbers                    | <a href="#">CF359-D2-D</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| Nuts for nuts                      | <a href="#">UVA 10944</a>                         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| Probability                        | <a href="#">UVA 11346</a>                         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
|                                    | <a href="#">SRM470-D1-500</a>                     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    | <a href="#">SPOJ COCONUTS</a>                     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
|                                    | <a href="#">CF592-D2-D</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    | <a href="#">UVA 1218</a>                          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
|                                    | <a href="#">SPOJ IOPC1207</a>                     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
|                                    | <a href="#">CF867-D12-E</a>                       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    |                                                   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| AlgoRace                           | <a href="#">CF189-D2-D</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| Modular Arithmet                   | <a href="#">CF604-D2-D</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| Lucky Transform                    | <a href="#">CF122-D2-D</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| Boring Partition                   | <a href="#">CF239-D2-D</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol. Find proof (See editorial comments)</a>                             |
| Spongebob and St                   | <a href="#">CF599-D2-D</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| How many trees?                    | <a href="#">CF9-D2-D</a>                          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    | <a href="#">CF1043-D12-E</a>                      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    | <a href="#">UVA 10982</a>                         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
|                                    | <a href="#">CF1060-D12-D</a>                      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| Chulhu                             | <a href="#">CF 104-D2-C</a>                       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| Anya and Ghosts                    | <a href="#">CF 308-D2-C</a>                       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| Square Subsets                     | <a href="#">CF 368-D2-C</a>                       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    |                                                   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| Angry Programme                    | <a href="#">UVA 11506</a>                         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| The New Rule in E                  | <a href="#">UVA 10742</a>                         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| Multiples of 3                     | <a href="#">SPOJ MULTQ3</a>                       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| <a href="#">TimeTravellingSale</a> | <a href="#">SRM492-D2-1000</a>                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| March of the Peng                  | <a href="#">UVA 12125</a>                         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| <a href="#">PeopleYouMayKnow</a>   | <a href="#">SRM447-D1-500</a>                     |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Don't use DP. Check it later in editorial. Sol</a>                       |
| The Game of 31                     | <a href="#">UVA 10578</a>                         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| Can you answer th                  | <a href="#">SPOJ GSS4</a>                         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| Area                               | <a href="#">TJU 1011</a>                          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| Volatile Kite                      | <a href="#">CF801-D2-D</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| Antifloyd                          | <a href="#">UVA 10987</a>                         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| Messenger                          | <a href="#">CF631-D2-D</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    | <a href="#">SRM144-D1-500</a>                     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    | <a href="#">SRM509-D1-500</a>                     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    | <a href="#">CF280-D1-C</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    | <a href="#">CF110-D2-D</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    | <a href="#">CF163-D12-C</a>                       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    | <a href="#">CF455-D1-B</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    |                                                   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| Infinite Maze                      | <a href="#">CF197-D2-D</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| Jeff and Furik                     | <a href="#">CF352-D2-D</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| Sagheer and Kind                   | <a href="#">CF812-D2-D</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                                                  |
| Dispute                            | <a href="#">CF242-D2-D</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
| Remainders Game                    | <a href="#">CF688-D2-D</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    | <a href="#">CF1075-D2-D</a>                       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    | <a href="#">CF1033-D12-D</a>                      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    | <a href="#">CF442-D1-B</a>                        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    | <a href="#">CF1025-D2-D</a>                       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |
|                                    | <a href="#">CF1072-D2-D</a>                       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                                                      |









| ff                  | Problem Code         | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments about your approach is interesting? | Mostafa Category                                | Category Code | Level | Quality |
|---------------------|----------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|-------------------------------------------------------|-------------------------------------------------|---------------|-------|---------|
|                     | AC Averages =>       | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                                     |                                                 |               |       |         |
|                     | CF101294-GYM-I       |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | dp, dp_ranges                                   | 32            | 4.5   | p1      |
|                     | SRM441-D1-250        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | dp, dp_ranges, [consecutive ranges, cyclic p    | 32            | 5     | p2      |
|                     | SRM536-D2-1000       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | dp, dp_ranges, [consecutive ranges]             | 32            | 5     | p1      |
| MessageMess         | SRM149-D1-500        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | dp, dp_ranges, impl, [consecutive ranges]       | 32            | 5     |         |
|                     | SRM555-D2-1000       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | dp, dp_ranges, [consecutive ranges]             | 32            | 5     |         |
|                     | SRM558-D1-250        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | dp, dp_ranges, [consecutive ranges] or bf       | 32            | 5.5   | p2      |
| Exploring Pyramid   | UVA 1362             |        |              |                 |                  |                |               | 0             |                   |              |          | Video Solution - Eng Ayman Salah                      | dp, dp_ranges                                   | 32            | 5.5   |         |
| Brackets sequence   | UVA 1626             |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | dp, dp_ranges                                   | 32            | 5.5   |         |
|                     | SRM509-D1-500        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | dp, dp_ranges, floyd, [cases]                   | 32            | 6     | p4      |
|                     | UVA 507              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | dp, dp_subrectangle, 1d, [more direct uva       | 36            | 3     |         |
|                     | UVA 10667            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | dp, dp_subrectangle, 2d                         | 36            | 3     |         |
| Big Maximum Sum     | CF75-D2-D            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | dp, dp_subrectangle, 2d, [actually greedy v     | 36            | 5     | p2      |
|                     | SPOJ FISHES          |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | dp, dp_subrectangle, 2d, observations, dot      | 36            | 5.5   | p3      |
| Reberland Linguist  | CF667-D2-C           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | dp, dp_table                                    | 37            | 4.5   | p3      |
| Red-Green Tower     | CF478-D2-D           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | dp, dp_table, dp_roll                           | 37            | 5     | p3      |
| Cunning Genie       | CF418-D1-B           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | dp, dp_table, dp_roll, dp_bitmasks, sortings    | 37            | 5.5   | p4      |
|                     | ZOJ 3305             |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | dp, dp_table or dp_bitmasks, all submasks       | 37            | 5.5   | p4      |
| An overnight danc   | CF814-D2-D           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | dp, dp_trees, geometry or greedy                | 38            | 5     | p3      |
|                     | CF161-D12-D          |        |              |                 |                  |                |               | 0             |                   |              |          | Reading: DP on Trees                                  | dp, dp_trees or dsu-on-trees                    | 38            | 5     | p2      |
| Vertex Cover        | SPOJ PT07X           |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | dp, dp_trees                                    | 38            | 5     |         |
|                     | CF337-D2-D           |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | dp, dp_trees or diameter like, [tricky to gue   | 38            | 5.5   | p4      |
| Chloe and please    | CF743-D2-D           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | dp, dp_trees                                    | 38            | 5.5   | p2      |
|                     | Timus 1362           |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | dp, dp_trees or greedy                          | 38            | 5.5   | p2      |
|                     | UVA 1218             |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | dp, dp_trees, [vertex cover related]            | 38            | 5.75  | p2      |
| Playing Cubes       | CF257-D2-B           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | game theory, greedy                             | 41            | 2.5   |         |
| Euclid's Game       | UVA 10368            |        |              |                 |                  |                |               | 0             |                   |              |          | Video Solution - Eng Moaz Rashad                      | game theory, gcd, dfs or pattern, [why each     | 41            | 3.5   | p2      |
|                     | CF1220-D12-C         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | game theory, adhoc                              | 41            | 3.5   | p2      |
| Alice and Bob       | CF347-D2-C           |        |              |                 |                  |                |               | 0             |                   |              |          | Video Solution - Eng Mohamed Nasser                   | game theory, gcd                                | 41            | 4     | p1      |
| Win or Freeze       | CF151-D2-C           |        |              |                 |                  |                |               | 0             |                   |              |          | Video Solution - Dr Mostafa Saad                      | game theory, divisors, greedy                   | 41            | 4     | p1      |
| Brownie Points      | UVA 10865            |        |              |                 |                  |                |               | 0             |                   |              |          | Video Solution - Eng Magdy Hasan                      | geometry                                        | 45            | 2     | p1      |
|                     | SRM436-D2-500        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, [slopes comparison]                   | 45            | 3     | p1      |
| Points in Figures:  | UVA 476              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry                                        | 45            | 3     |         |
| Watering Flowers    | CF617-D2-C           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, bf                                    | 45            | 3     |         |
| Pouring Rain        | CF667-D2-A           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, physics                               | 45            | 3     |         |
| Fourth Point !!     | UVA 10242            |        |              |                 |                  |                |               | 0             |                   |              |          | Video Solution - Eng Magdy Hasan                      | geometry, vectors addition                      | 45            | 3     |         |
| Captain Marmot      | CF474-D2-C           |        |              |                 |                  |                |               | 0             |                   |              |          | Video Solution - Dr Mostafa Saad                      | geometry, check square, point rotation, bf      | 45            | 3.5   | p2      |
| Overlapping Recta   | UVA 460              |        |              |                 |                  |                |               | 0             |                   |              |          | Video Solution - Eng Muntaser Abukadeja               | geometry                                        | 45            | 3.5   |         |
| Xrange's Pancake    | HACKR xrange-and-piz |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, adhoc                                 | 45            | 4     | p2      |
|                     | HACKR a-circle-and-a |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, ccw, parametric equ, in circle        | 45            | 4     | p2      |
|                     | SPOJ FACENEMY        |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, angles, precision                     | 45            | 4     | p1      |
| k-Multiple Free Se  | CF275-D2-C           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry                                        | 45            | 4     |         |
| Gerald's Hexagon    | CF560-D2-C           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry                                        | 45            | 4     |         |
| View Angle          | CF257-D2-C           |        |              |                 |                  |                |               | 0             |                   |              |          | Editorial - Eng Ahmed Osama                           | geometry, angles                                | 45            | 4     |         |
| Watchmen            | CF651-D2-C           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, datastructures                        | 45            | 4     |         |
| Bicycle Race        | CF659-D2-D           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, impl, [very nice, o(1) and o(n) sol   | 45            | 4.5   | p3      |
| Pyramids            | SPOJ PIR             |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, formula or matrix determinant         | 45            | 4.5   | p1      |
| Pythagorean Triplet | CF707-D2-C           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, triangles, formula                    | 45            | 4.5   | p2      |
|                     | SPOJ BILLIARD        |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, angles, physics                       | 45            | 5     |         |
| Cupboard and Ball   | CF342-D2-C           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry                                        | 45            | 5     | p3      |
|                     | CF1064-D2-E          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, binary search, interactive            | 45            | 5     | p3      |
|                     | CF961-D12-D          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry                                        | 45            | 5     | p2      |
|                     | CF101917-D12-E       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, [lpl scared in contest, but easy]     | 45            | 5     | p2      |
|                     | CF552-D2-D           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, bf, counting, treemaps                | 45            | 5     | p2      |
|                     | CF1016-D2-E          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, binary search                         | 45            | 5     | p2      |
|                     | CF1058-D2-D          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, triangles, number theory              | 45            | 5     | p2      |
|                     | UVA 1342             |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, plane graph                           | 45            | 5     |         |
|                     | CF101864-GYM-L       |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, binary search or bf, greedy           | 45            | 5.5   | p3      |
|                     | CF80-D2-D            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, probability or algebra                | 45            | 5.5   | p2      |
|                     | UVA 11648            |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, trapezoid formula, binary search      | 45            | 6     | p2      |
|                     | UVA 1333             |        |              |                 |                  |                |               | 0             |                   |              |          | Sol - Text/Background Clarification                   | geometry, triangles, angles, parallelogram l    | 45            | 6     | p1      |
| Hit Ball            | CF203-D2-D           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, 3d, impl, math, [physics, kinemat     | 46            | 5     | p2      |
|                     | UVA 453              |        |              |                 |                  |                |               | 0             |                   |              |          | Learn Handling Precisions                             | geometry, circles, [direct circle intersection, | 47            | 2     |         |
| Wifi Access         | UVA 12748            |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, circles, distances                    | 47            | 2     |         |
| Rings and Glue      | UVA 10301            |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, circles, dsu                          | 47            | 3     | p1      |
| Square Pegs And     | UVA 356              |        |              |                 |                  |                |               | 0             |                   |              |          | Sol to read                                           | geometry, circles                               | 47            | 3     |         |
| The Circumferenc    | UVA 438              |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, circles                               | 47            | 3     |         |
| Points in Figures:  | UVA 477              |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, circles                               | 47            | 3.5   |         |
| Special Olympics    | CF199-D2-B           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, circles, impl                         | 47            | 4     |         |
| Biathlon            | CF84-D2-C            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, circles, impl                         | 47            | 4     |         |
| Packing polygons    | UVA 10005            |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, circles, polygon, [polygon inside p   | 47            | 5     | p4      |
|                     | SRM473-D1-500        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, circles, triangles, thales' theorem   | 47            | 5     | p3      |
|                     | SPOJ ALIENS          |        |              |                 |                  |                |               | 0             |                   |              |          | Sol - Practice on min enclosing circle                | geometry, circles, min enclosing circle, [=sp   | 47            | 5     | p2      |
|                     | CF1059-D2-D          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, circles, binary search                | 47            | 5.25  | p3      |
|                     | HACKR house-location |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, circles, algebra, impl                | 47            | 5.5   | p3      |
|                     | UVA 10180            |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, circles, tangents, point on segme     | 47            | 5.5   | p2      |
| Railway             | UVA 10263            |        |              |                 |                  |                |               | 0             |                   |              |          | Sol to read                                           | geometry, lines, distances, [=uva 460]          | 48            | 3     | p3      |
| Lining Up           | UVA 270              |        |              |                 |                  |                |               | 0             |                   |              |          | Video Solution - Eng Mohamed Nasser, Don't Co         | geometry, lines, line up                        | 48            | 3     | p3      |
| Campus Roads        | UVA 11473            |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, lines, distances, impl                | 48            | 3     | p2      |
| Polyline            | CF617-D2-D           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, lines, impl                           | 48            | 3     |         |
| Jack Straws         | UVA 273              |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, lines, intersection, shortest path    | 48            | 3     |         |
| Isolated Segments   | UVA 11343            |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, lines, intersections                  | 48            | 3     |         |
| Intersecting Lines  | UVA 378              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, lines                                 | 48            | 3.5   |         |
|                     | SRM373-D2-1000       |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, lines, lines intersection, rectangle  | 48            | 4     |         |
| Intersecting Line S | UVA 866              |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, lines, intersections                  | 48            | 4     |         |
|                     | SRM368-D1-500        |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, lines, polyline intersection, bf, na  | 48            | 4     |         |
| Gleaming the Cub    | UVA 737              |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, lines, intersections                  | 48            | 4     |         |
| Water Falls         | UVA 833              |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, lines, distances, adhoc               | 48            | 4     | p3      |
| How Many Points     | UVA 10790            |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, lines, intersections, counting, for   | 48            | 4     | p1      |
| River Crossing      | UVA 10514            |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, lines, distances, floyd               | 48            | 5     |         |
| BestTriangulation   | SRM278-D2-500        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, polygon, area, [just triangle area    | 49            | 2     |         |
| Triangle            | CF408-D2-C           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, polygon                               | 49            | 4     |         |
|                     | UVA 11665            |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, polygon, pip, polygons intersect      | 49            | 4     |         |
|                     | TIMUS 1599           |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, polygon, pip, winding numbers, [l     | 49            | 4.5   | p1      |
|                     | UVA 881              |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, polygon, pip, polygons inside pol     | 49            | 4.5   |         |
|                     | CF340-D2-B           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, polygon, bf                           | 49            | 5     | p2      |
| Volatile Kite       | CF801-D2-D           |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, polygon, binary search                | 49            | 5     | p2      |
| Polygons            | UVA 137              |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, polygon, pip, intersections or cor    | 49            | 5.5   | p3      |
| Area                | TJU 1011             |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | geometry, polygon, pick's theorem               | 52            | 4.5   | p1      |
| Trees on My Islan   | UVA 10088            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | geometry, polygon, pick's theorem, gcd          | 52            | 5     |         |
|                     | LIVEARCHIVE 2831     |        |              |                 |                  |                |               | 0             |                   |              |          | Use polygon cut                                       | geometry, polygon, polygon cut                  | 53            | 4     |         |
| Video Surveillance  | UVA 588              |        |              |                 |                  |                |               | 0             |                   |              |          | Use polygon cut                                       | geometry, polygon, polygon cut or adhoc         | 53            | 6     | p5      |
| The Skyline Proble  | UVA 105              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | greedy, geometry                                |               | 3     |         |
| Marcus              | UVA 10452            |        |              |                 |                  |                |               | 0             |                   |              |          | Video Solution - Eng Ayman Salah                      | graph                                           | 55            | 3     |         |
| Trees on the level  | UVA 122              |        |              |                 |                  |                |               | 0             |                   |              |          | Video Solution - SolverToBe (Java)                    | graph, trees                                    | 55            | 3     |         |
| PT07Z               | SPOJ PT07Z           |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | graph, tree diameter                            | 55            | 3     |         |
| Roads in the North  | UVA 10308            |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | graph, tree diameter                            | 55            | 3     |         |
|                     | CF1068-D2-C          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | graph, adhoc                                    | 55            | 4     | p2      |

| ff                  | Problem Code     | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments about your approach is interesting?  | Mostafa Category                                 | Category Code | Level | Quality |
|---------------------|------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|--------------------------------------------------------|--------------------------------------------------|---------------|-------|---------|
|                     | AC Averages ==>  | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                                      |                                                  |               |       |         |
| Eternal Victory     | CF61-D2-D        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, greedy                                    | 55            | 4     | p2      |
| Is It A Tree?       | UVA 615          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, trees                                     | 55            | 4     | p1      |
| Mahmoud and Elh     | CF959-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Salah</a>     | graph, trees, constructive                       | 55            | 4     |         |
| Central Post Office | UVA 12379        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, tree diameter                             | 55            | 4     |         |
| The Tree Root       | UVA 10459        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, tree diameter                             | 55            | 4.5   | p3      |
| Xor-tree            | CF430-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, bf                                        | 55            | 5     |         |
| Renting Bikes       | CF363-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, cycle, greedy                             | 55            | 5     |         |
| Regular Bridge      | CF550-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, prove using e.g. soc                      | 55            | 5     |         |
|                     | CF486-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, trees, dfs, prefix sum or dp_trees        | 55            | 5.5   | p5      |
| Cycles              | CF233-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, cycle                                     | 55            | 5.5   | p3      |
|                     | CF459-D2-E       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dp, sortings                              | 55            | 5.5   | p3      |
|                     | CF1060-D12-D     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, greedy                                    | 55            | 5.5   | p3      |
|                     | UVA 10982        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, greedy, [close to max cut]                | 55            | 5.5   | p3      |
|                     | CF592-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, tree diameter                             | 55            | 5.5   | p3      |
| BITMAP - Bitmap     | SPOJ BITMAP      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, bfs, multiscr, multidest                  | 57            | 3     | p3      |
| Pouring water       | SPOJ POUR1       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Moaz Rashad</a>       | graph, bfs                                       | 57            | 3     |         |
| Jugs                | UVA 571          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>       | graph, bfs                                       | 57            | 4     | p1      |
| Tic-Tac-Toe ( I )   | SPOJ TOE1        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ayman Salah</a>       | graph, bfs                                       | 57            | 4     |         |
| Tic-Tac-Toe ( II )  | SPOJ TOE2        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Essam AlNaggar</a>    | graph, bfs                                       | 57            | 4     |         |
| Knight Moves        | UVA 439          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Magdy Hasan</a>       | graph, bfs, chess or dfs                         | 57            | 4     |         |
| King's Path         | CF242-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>       | graph, bfs                                       | 57            | 4.5   |         |
| Theseus and labyr   | CF676-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, bfs, impl                                 | 57            | 4.5   | p2      |
| Wandering Queen     | SPOJ QUEEN       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol to read</a>                            | graph, bfs                                       | 57            | 4.5   | p1      |
| Restore Graph       | CF404-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, bfs                                       | 57            | 4.5   |         |
| Key Task            | SPOJ CERC07K     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, bfs, bitmask                              | 57            | 4.5   |         |
| Cleaning Robot      | SPOJ CLEANRBT    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, bfs, bitmask or bfs preprocess then       | 57            | 4.5   |         |
|                     | UVA 10888        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, bfs, dp or weighted matching              | 57            | 5     | p3      |
| Text Editor         | CF253-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, bfs or greedy, [search in 2d grid]        | 57            | 5     | p2      |
| Tobo or not Tobo    | SPOJ ANARC08A    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, bfs, trie, hashing or meet in middle      | 57            | 5     |         |
|                     | CF1005-D3-F      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, bfs                                       | 57            | 5.25  | p2      |
|                     | TIMUS 1498       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, bfs, [chess, tricky cases]                | 57            | 5.5   | p2      |
|                     | UVA 11573        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Learn 0/1 BFS</a>                          | graph, bfs, 0/1 bfs, [-spo] kathi]               | 57            | 5.5   | p2      |
|                     | CF787-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, bfs, cyclic games                         | 57            | 5.5   | p1      |
|                     | CF811-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, bfs, interactive                          | 57            | 6     | p2      |
|                     | UVA 10461        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs, [finish computation times]           | 60            | 3     | p1      |
| Roads in Berland    | CF25-D2-C        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs                                       | 60            | 4     | p2      |
| Party               | CF116-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs                                       | 60            | 4     | p1      |
| Forming Teams       | CF216-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>       | graph, dfs                                       | 60            | 4     |         |
| Block Tower         | CF327-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs                                       | 60            | 4     |         |
| Soldier and Cards   | CF546-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs                                       | 60            | 4     |         |
| Kefa and Park       | CF580-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>   | graph, dfs                                       | 60            | 4     | p2      |
| Maze                | CF378-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs, [reverse thinking]                   | 60            | 4.5   | p1      |
| Exchange Rates      | UVA 10113        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs, impl                                 | 60            | 4.5   | p1      |
| Ice Cave            | CF540-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs                                       | 60            | 4.5   |         |
| Ordering            | UVA 872          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs                                       | 60            | 4.5   |         |
| Directed Roads      | CF711-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs, combinatorics, formula               | 60            | 5     | p3      |
|                     | SPOJ BIA         |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, dfs or directed articulation points alg   | 60            | 5     | p2      |
| Choosing Capital f  | CF219-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs or dp_trees                           | 60            | 5     |         |
|                     | CF1075-D2-D      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs, interactive                          | 60            | 5.5   | p3      |
| Infinite Maze       | CF197-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs                                       | 60            | 5.5   |         |
| Cycle in Graph      | CF263-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs                                       | 60            | 5.5   |         |
| T-decomposition     | CF237-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs, greedy                               | 60            | 5.5   |         |
| Robbery             | UVA 707          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, dfs or dp                                 | 60            | 5.75  |         |
| Persistent Bookca   | CF707-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, dfs, bitset or persistent segment tree    | 60            | 6     | p3      |
| Moodular Arithmet   | CF604-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, dfs, fermat, [rearrangement property]     | 60            | 6     | p2      |
| The Seasonal War    | UVA 352          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mohamed Nasser</a>    | graph, dfs, flood-fill                           | 61            | 2     |         |
| Battleships         | UVA 11953        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Aya Elymany</a>       | graph, dfs, flood-fill                           | 61            | 3.5   |         |
| Maze Exploration    | UVA 784          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Mahmoud Adel</a>      | graph, dfs, flood-fill                           | 61            | 3.5   |         |
| Continents          | UVA 11094        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Ayman Salah</a>       | graph, dfs, flood-fill                           | 61            | 4     |         |
|                     | SRM297-D1-500    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs, flood-fill or bfs, bf                | 61            | 5     | p3      |
| Equivalent Strings  | CF560-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol to learn</a>                           | graph, dfs, isomorphism or d&c, hashing          | 62            | 4     | p2      |
| Subway tree syste   | LIVEARCHIVE 2935 |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, dfs, isomorphism, canonical form or       | 62            | 4.5   | p4      |
| Hierarchy           | SPOJ MAKETREE    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Yahia Ashraf</a>      | graph, dfs, topological sort                     | 63            | 2     |         |
| Ordering Tasks      | UVA 10305        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Yahia Ashraf</a>      | graph, dfs, topological sort                     | 63            | 3     |         |
|                     | SRM419-D2-1000   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs, topological sort, cycles             | 63            | 4     |         |
| Spreadsheet         | UVA 196          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs, topological sort or dp               | 63            | 4     | p3      |
| Rankings            | UVA 12263        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, dfs, topological sort                     | 63            | 4     | p2      |
| Pick up sticks      | UVA 11686        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, dfs, topological sort, detect cycles      | 63            | 4     |         |
|                     | SRM550-D2-1000   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs, topological sort                     | 63            | 5     | p3      |
| Robot Rapping Re    | CF645-D12-D      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs, topological sort, binary search      | 63            | 5     | p3      |
| Gifts by the List   | CF681-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs, topological sort, impl               | 63            | 5     | p2      |
| Sagheer and Kind    | CF812-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, dfs, topological sort or euler, [https:// | 63            | 6     | p4      |
| Shopping            | SPOJ SHOP        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dfs, topological sort                     | 64            | 3     |         |
| Sending email       | UVA 10986        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dijkstra                                  | 64            | 3     |         |
| MELE3               | SPOJ MELE3       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, dijkstra                                  | 64            | 3     |         |
| Roads               | SPOJ ROADS       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, dijkstra or dp                            | 64            | 4.5   | p3      |
| Lift Hopping        | UVA 10801        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dijkstra                                  | 64            | 4.5   |         |
|                     | UVA 10740        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, dijkstra, kth sp. [k <= 10]               | 64            | 5     | p3      |
| Volleyball          | CF96-D2-D        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dijkstra, 2 dijkstra                      | 64            | 5     | p3      |
|                     | UVA 12047        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, dijkstra                                  | 64            | 5.5   | p3      |
|                     | UVA 10342        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol - read the statement clarification</a> | graph, dijkstra, kth sp (k=2) or floyd           | 64            | 5.5   | p3      |
| Hotel booking       | UVA 11635        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, dijkstra                                  | 64            | 5.5   |         |
| IP-TV               | UVA 1174         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dsu                                       | 65            | 2     |         |
| Count the Faces.    | UVA 10178        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Read first Euler Formula</a>               | graph, dsu or dfs, cycles                        | 65            | 4     | p2      |
| Learning Language   | CF278-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dsu                                       | 65            | 4     |         |
| Virtual Friends     | UVA 11503        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Eng Moaz Rashad</a>       | graph, dsu                                       | 65            | 4     |         |
| Almost Union-Find   | UVA 11987        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, dsu                                       | 65            | 4.5   | p3      |
| Cthulhu             | CF104-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dsu                                       | 65            | 4.5   |         |
| The Child and Zoo   | CF437-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dsu                                       | 65            | 5     |         |
| Mahmoud and a D     | CF766-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Solver to be (Java)</a>   | graph, dsu, [offline processing]                 | 65            | 5     | p3      |
|                     | CF1012-D1-B      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dsu                                       | 65            | 5.25  | p2      |
|                     | UVA 12128        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dsu, dijkstra like or binary search, bf   | 65            | 5.5   | p2      |
| Connected Compco    | CF292-D12-D      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, dsu                                       | 65            | 5.75  | p3      |
| Trip Routing        | UVA 186          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, floyd, path print                         | 68            | 4     | p3      |
| Numbering Paths     | UVA 125          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, floyd, paths counting                     | 68            | 4.5   | p5      |
| Frogger             | UVA 534          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, floyd, minimax or dsu                     | 68            | 4.5   | p4      |
| Travel in Desert    | UVA 10816        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, floyd, binary search                      | 68            | 4.5   |         |
| Identifying Concur  | UVA 334          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, floyd                                     | 68            | 4.5   |         |
| Greg and Graph      | CF296-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, floyd                                     | 68            | 5     | p2      |
| Dima and Bacteria   | CF400-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, floyd, dfs                                | 68            | 5     | p2      |
| AlgoRace            | CF189-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, floyd                                     | 68            | 5.25  | p4      |
| Antifloyd           | UVA 10987        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, floyd, antifloyd                          | 68            | 5.5   | p4      |
| Unique World        | UVA 10448        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Video Solution - Dr Mostafa Saad</a>       | graph, floyd, dp                                 | 68            | 5.5   | p2      |
| Arbitrage           | UVA 104          |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, floyd                                     | 68            | 6.25  | p2      |
| Potholers           | SPOJ POTTHOLE    |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, max-flow                                  | 71            | 3     |         |
| Power Transmissk    | UVA 10330        |        |              |                 |                  |                |               | 0             |                   |              |          | <a href="#">Sol</a>                                    | graph, max-flow, vertex constraints              | 71            | 4     |         |
| The Problem with t  | UVA 10092        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                        | graph, max-flow, [direct bipartite is slow]      | 71            | 4.5   | p3      |





| ff                           | Problem Code     | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category                                        | 1-2 line Comments about your approach is interesting? | Mostafa Category                                         | Category Code | Level | Quality |
|------------------------------|------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|-------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------|---------------|-------|---------|
|                              | AC Averages =>   | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0                                               | 0                                                     |                                                          |               |       |         |
| Tanya and Postcard           | CF518-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 2     |         |
| Mike and Fun                 | CF548-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 2     |         |
| Covered Path                 | CF534-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 2     |         |
| Print Check                  | CF631-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 2     |         |
| Lucky Mask                   | CF146-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 2     |         |
| Special Offer! Super         | CF219-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 2     | p2      |
| Non-square Equation          | CF233-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 2     |         |
| Flag Day                     | CF357-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 2     |         |
| Sereja and Mirrors           | CF426-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 2     |         |
| Little Pony and Socks        | CF454-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 2     |         |
| MUH and Importers            | CF471-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 2     |         |
| Gena's Code                  | CF614-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 2     |         |
| Opposites Attract            | CF131-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 2     |         |
| Little Pigs and Wolves       | CF116-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 2     |         |
| Cosmic Tables                | CF222-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 2     |         |
| Prime Matrix                 | CF271-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 2     |         |
| Wet Shark and Bits           | CF621-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 | Video Solution - Eng Mahmoud Mabrok                   | impl                                                     | 86            | 2     |         |
|                              | CF1030-D12-B     |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl, math                                               | 86            | 2     |         |
| Facetook Priority Queue      | CF75-D2-B        |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl, sorting                                            | 86            | 2     |         |
|                              | CF490-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | graph, constructive, adhoc                               | 86            | 3     | p2      |
| Hanoi Tower                  | TIMUS 1054       |        |              |                 |                  |                |               | 0             |                   |              | Sol                                             |                                                       | impl, recursion, tower of hanoi                          | 86            | 3     | p2      |
| Treasure                     | CF495-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 4     |         |
| Game                         | CF69-D2-C        |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 4     |         |
| Accordian Patience           | UVA 127          |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Eng Moaz Rashad                |                                                       | impl                                                     | 86            | 4     | p1      |
| Beautiful Sets of Pairs      | CF268-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl, constructive                                       | 86            | 4     |         |
| Appleman and Tree            | CF462-D2-C       |        |              |                 |                  |                |               | 0             |                   |              | Sol                                             |                                                       | impl, sorting, huffman coding                            | 86            | 4     |         |
| Three Logos                  | CF581-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl                                                     | 86            | 4.5   | p2      |
| Guess Your Way Out           | CF507-D2-C       |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Dr Mostafa Saad                |                                                       | impl, math                                               | 86            | 4.5   | p2      |
|                              | CF1042-D12-D     |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl or segment tree or bit                              | 86            | 5     | p2      |
| Mafia                        | CF349-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl, math                                               | 86            | 5     | p2      |
| Unusual Product              | CF405-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl, math, [symbolic thinking]                          | 86            | 5     | p1      |
|                              | CF101187-GYM-F   |        |              |                 |                  |                |               | 0             |                   |              | Sol                                             |                                                       | impl                                                     | 86            | 5.25  | p2      |
| Special Grid                 | CF435-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | impl, greedy                                             | 86            | 5.5   | p2      |
| Theatre Square               | CF1-D12-A        |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Solver to be (Java)            |                                                       | math                                                     | 87            | 1.5   |         |
|                              | CF1204-D2-A      |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Dr Mostafa Saad                |                                                       | math, log, binary, pattern observation                   | 87            | 2     | p2      |
| Balanced Rating Contest      | CF1237-D12-A     |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, analysis                                           | 87            | 2     | p3      |
| The Drunk Jailer             | LIVEARCHIVE 2557 |        |              |                 |                  |                |               | 0             |                   |              | Find a formula                                  |                                                       | math or bf                                               | 87            | 2     | p1      |
| Product                      | UVA 10106        |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Eng Youssef El Ghareeb_Don't   |                                                       | math                                                     | 87            | 2     |         |
| To Carry or not to Carry     | UVA 10469        |        |              |                 |                  |                |               | 0             |                   |              | Sol                                             |                                                       | math                                                     | 87            | 2     |         |
| Adding Reversed Integers     | UVA 713          |        |              |                 |                  |                |               | 0             |                   |              | Don't use big integer class. Write simple array |                                                       | math                                                     | 87            | 2     |         |
| Dreamoon and Whiskers        | CF476-D2-B       |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Eng Mohamed Adel               |                                                       | math, combinatorics, bitmasks                            | 87            | 2     | p3      |
|                              | CF1051-D2-B      |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math                                                     | 87            | 2.5   |         |
| Escape                       | CF148-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math                                                     | 87            | 2.5   |         |
| Restoring Painting           | CF675-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math                                                     | 87            | 2.5   |         |
| Caisa and Pylons             | CF463-D2-B       |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Eng Muntaser Abukadeja         |                                                       | math, impl                                               | 87            | 2.5   |         |
| T-primes                     | CF230-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, number theory                                      | 87            | 2.5   |         |
|                              | CODECHEF_GCDMOD  |        |              |                 |                  |                |               | 0             |                   |              | Sol uses __int128 to avoid overflow             |                                                       | math, __int128                                           | 87            | 3     | p3      |
| Polycarpus' Dilemma          | CF534-D2-C       |        |              |                 |                  |                |               | 0             |                   |              | Sol                                             |                                                       | math, greedy, careful impl                               | 87            | 3     | p3      |
|                              | CF1059-D2-C      |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, constructive                                       | 87            | 4     | p3      |
| Number Sequence              | UVA 10706        |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math                                                     | 87            | 4     | p2      |
| Divisible by Seven           | CF376-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, number theory                                      | 87            | 4     | p2      |
| Fractions Again?!            | UVA 10976        |        |              |                 |                  |                |               | 0             |                   |              | Sol to read                                     |                                                       | math, number theory                                      | 87            | 4     | p1      |
| Plant                        | CF186-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math                                                     | 87            | 4     |         |
| Magic Formulas               | CF424-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math                                                     | 87            | 4     |         |
| Duff in Love                 | CF588-D2-B       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math                                                     | 87            | 4     |         |
| Light, more light            | UVA 10110        |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Eng Amr Saud                   |                                                       | math                                                     | 87            | 4     |         |
| Power of Cryptography        | UVA 113          |        |              |                 |                  |                |               | 0             |                   |              | Sol to read                                     |                                                       | math, log, [double limits]                               | 87            | 4     | p3      |
| Round Table Knight           | CF71-D2-C        |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math or dp                                               | 87            | 4     |         |
| Lucky Permutation            | CF304-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, constructive                                       | 87            | 4     |         |
| Vasya and Petya's Game       | CF577-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, impl                                               | 87            | 4     | p2      |
|                              | CF1239-D1-A      |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, pattern                                            | 87            | 4.25  | p3      |
| The ? 1 ? 2 ? ... ?          | UVA 10025        |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math or binary search                                    | 87            | 4.5   | p2      |
| Secrets                      | CF334-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math                                                     | 87            | 4.5   |         |
| The Meaningless (Contest)    | CF834-D2-C       |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Solver to be (Java)            |                                                       | math                                                     | 87            | 4.5   |         |
| Find Maximum                 | CF353-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, bits                                               | 87            | 4.5   |         |
| Plus and Square Root         | CF716-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, constructive                                       | 87            | 4.5   |         |
| Bear and Prime 10            | CF680-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, constructive, interactive                          | 87            | 4.5   |         |
|                              | CF1040-D2-D      |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, randomization, binary search, interaction          | 87            | 5     | p4      |
| Count Good Substrings        | CF451-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, adhoc, palindromes, [short code]                   | 87            | 5     | p3      |
| Tavas and Karafs             | CF535-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, binary search                                      | 87            | 5     | p2      |
| As Fast As Possible          | CF701-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, binary search, precision                           | 87            | 5     | p2      |
|                              | CF955-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, number theory                                      | 87            | 5     | p2      |
|                              | CF45-D12-D       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, randomization                                      | 87            | 5     | p2      |
| Ciel and Robot               | CF322-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, impl, [cases]                                      | 87            | 5     | p1      |
| Crazy Town                   | CF499-D2-C       |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Dr Mostafa Saad                |                                                       | math, number theory, greedy                              | 87            | 5     | p1      |
| About Bacteria               | CF199-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math                                                     | 87            | 5     |         |
| DNA Alignment                | CF520-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math                                                     | 87            | 5     |         |
| Predict Outcome of Game      | CF451-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, equations, impl                                    | 87            | 5     | p2      |
| Analyzing Polyline           | CF195-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, sortings                                           | 87            | 5     |         |
|                              | CF1016-D12-D     |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, xor, bitwise, constructive                         | 87            | 5.25  | p3      |
| Quantity of Strings          | CF151-D2-D       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, repeated squaring, graph                           | 87            | 5.5   | p4      |
| How many trees?              | CF9-D2-D         |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math or dp_tree                                          | 87            | 5.5   | p2      |
| The Errant Physicist         | UVA 128          |        |              |                 |                  |                |               | 0             |                   |              | Sol                                             |                                                       | math                                                     | 87            | 5.5   |         |
| Software CRC                 | UVA 128          |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Eng Moaz Rashad                |                                                       | math                                                     | 87            | 5.5   |         |
| Jeff and Furk                | CF352-D2-D       |        |              |                 |                  |                |               | 0             |                   |              | Sol                                             |                                                       | math or dp_expectation                                   | 87            | 6     | p3      |
| Magical Array                | CF84-D2-B        |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, combinatorics                                      | 89            | 2.5   |         |
| Chocolate                    | CF617-D2-B       |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Eng Yahia Ashraf               |                                                       | math, combinatorics                                      | 89            | 2.5   |         |
| Colorful Field               | CF79-D12-B       |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Solver to be (Java)            |                                                       | math, mod                                                | 89            | 2.5   | p3      |
| The World is a Theatre       | CF131-D2-C       |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Eng Youssef Ali                |                                                       | math, combinatorics                                      | 89            | 4     |         |
| Pocket Book                  | CF152-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, combinatorics                                      | 89            | 4     |         |
| Black and white photograph   | UVA 11231        |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Eng Amr Saud                   |                                                       | math, combinatorics, counting                            | 89            | 4     |         |
|                              | CF758-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, combinatorics                                      | 89            | 5     | p3      |
|                              | CF459-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, combinatorics, constructive                        | 89            | 5     | p3      |
|                              | HACKER_ajourney  |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, combinatorics, first/last k digits 2^n, [          | 89            | 5     | p3      |
| Shaass and Lights            | CF294-D2-C       |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Dr Mostafa Saad                |                                                       | math, combinatorics                                      | 89            | 5.5   | p4      |
|                              | CF869-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, combinatorics or dp_counting                       | 89            | 5.5   | p3      |
| Tourist Problem              | CF340-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, combinatorics, impl                                | 89            | 5.5   | p1      |
| Fox Dividing Cheesecake      | CF371-D2-B       |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Eng Abanob Ashraf              |                                                       | math, factorial                                          | 94            | 2.5   |         |
| Permalex                     | UVA 153          |        |              |                 |                  |                |               | 0             |                   |              | Sol                                             |                                                       | math, factorial, permutations, duplicates, factorization | 94            | 4.5   | p3      |
| Prime Factors                | UVA 583          |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, factorization                                      | 95            | 2     |         |
| Easy Number Characterization | CF236-D2-B       |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Eng Yahia Ashraf               |                                                       | math, factorization                                      | 95            | 3     |         |
| Mr. Azad and his Friends     | UVA 10490        |        |              |                 |                  |                |               | 0             |                   |              | Sol to read                                     |                                                       | math, factorization                                      | 95            | 3     |         |
| Prime Land                   | UVA 516          |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, factorization                                      | 95            | 3     |         |
| Perfect P-th Power           | UVA 10622        |        |              |                 |                  |                |               | 0             |                   |              | Video Solution - Eng Moaz Rashad                |                                                       | math, factorization                                      | 95            | 4     | p1      |
| Factovisors                  | UVA 10139        |        |              |                 |                  |                |               | 0             |                   |              | Sol to read                                     |                                                       | math, factorization, primes, [factorize x!]              | 95            | 4     |         |
|                              | CF1047-D2-C      |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, factorization                                      | 95            | 4.5   | p3      |
| DDF                          | UVA 547          |        |              |                 |                  |                |               | 0             |                   |              |                                                 |                                                       | math, factorization, divisors sum, multiview             | 95            | 4.5   |         |

| ff | Problem Code                    | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments about your approach is interesting? | Mostafa Category                                | Category Code | Level | Quality |
|----|---------------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|-------------------------------------------------------|-------------------------------------------------|---------------|-------|---------|
|    | AC Averages => UVA 10174        | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                                     | math, factorization, case analysis              | 95            | 5     |         |
|    | UVA 11347                       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, factorization, divisors sum               | 95            | 5     |         |
|    | CF1033-D12-D                    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, factorization                             | 95            | 5.5   | p3      |
|    | Remainders Game CF688-D2-D      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, factorization, gcd, lcm, observations     | 95            | 6     | p4      |
|    | Primitive Root SPOJ PROOT       |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, factorization, primitive roots            | 95            | 6.25  | p4      |
|    | UVA 12869                       |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, formula                                   | 98            | 5     | p2      |
|    | Combinations UVA 369            |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, gcd, comb formula                         | 99            | 2     |         |
|    | Pi UVA 412                      |        |              |                 |                  |                |               | 0             |                   |              |          | Video Solution - Eng Mohamed Adel                     | math, gcd                                       | 99            | 3     |         |
|    | Trains CF88-D2-C                |        |              |                 |                  |                |               | 0             |                   |              |          | Video Solution - Solver to be (Java)                  | math, gcd or adhoc                              | 99            | 4     |         |
|    | Mint UVA 10717                  |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, gcd, lcm                                  | 99            | 4     |         |
|    | The Big Race CF592-D2-C         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, gcd, lcm, [overflow]                      | 99            | 4.5   | p3      |
|    | LCM Cardinality UVA 10892       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, gcd, lcm                                  | 99            | 4.5   |         |
|    | Rational Resistance CF344-D2-C  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, gcd                                       | 99            | 5     | p3      |
|    | LCM Challenge CF236-D2-C        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, gcd, lcm                                  | 99            | 5     |         |
|    | CF1010-D1-C                     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, gcd, mod, number theory                   | 99            | 5.5   | p1      |
|    | AICoder026-AGC-B                |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, gcd, cases                                | 99            | 6     | p3      |
|    | SPOJ EASYMATH                   |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, inclusion-exclusion, lcm                  | 101           | 3     |         |
|    | Hamburgers CF371-D2-C           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, inclusion-exclusion, binary search        | 101           | 3     |         |
|    | The Lottery UVA 10325           |        |              |                 |                  |                |               | 0             |                   |              |          | Video Solution - Eng Amr Bahaa [4]                    | math, inclusion-exclusion, gcd, overflow        | 101           | 4     |         |
|    | CF101933-GYM-K                  |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, inclusion-exclusion                       | 101           | 4     |         |
|    | CF372-D1-B                      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, inclusion-exclusion                       | 101           | 5.75  | p2      |
|    | SPOJ MSKYCODE                   |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, inclusion-exclusion                       | 101           | 6     | p3      |
|    | CF101992-GYM-D                  |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, inclusion-exclusion                       | 101           | 6     | p3      |
|    | Equation UVA 727                |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, infix to postfix                          | 102           | 4     |         |
|    | Farm TIMUS 1349                 |        |              |                 |                  |                |               | 0             |                   |              |          | Know Fermat's Last Theorem (ignore proof)             | math, math_adhoc, fermat last theorem           | 104           | 2     | p2      |
|    | Summation of Poly UVA 10302     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, math_adhoc, polynomials                   | 104           | 2     |         |
|    | HACKR lower-3-colorin           |        |              |                 |                  |                |               | 0             |                   |              |          | Learn Fermat's little theorem                         | math, math_adhoc, fermat little theorem         | 104           | 3     | p1      |
|    | R U Kidding Mr. Fr UVA 10509    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, math_adhoc, patterns                      | 104           | 3.5   |         |
|    | Polly the Polynomi UVA 498      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, math_adhoc, polynomials                   | 104           | 3.5   |         |
|    | Jzzhu and Sequen CF450-D2-B     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, matrix, matrix exponent                   | 105           | 2.5   |         |
|    | Mirror, Mirror UVA 466          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, matrix, rotate, reflect, impl             | 105           | 3     | p1      |
|    | Clear Symmetry CF202-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, matrix, bf                                | 105           | 4     |         |
|    | End of Fun SPOJ DCEPC12E        |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, matrix                                    | 105           | 4.5   |         |
|    | Uniform Generator UVA 408       |        |              |                 |                  |                |               | 0             |                   |              |          | Video Solution - Eng Yahia Ashraf                     | math, mod                                       | 109           | 3     |         |
|    | Be Efficient UVA 11155          |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, mod                                       | 109           | 5     |         |
|    | Quiz CF337-D2-C                 |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, mod, pow, greedy                          | 109           | 5.5   | p3      |
|    | Cows and Cars UVA 10491         |        |              |                 |                  |                |               | 0             |                   |              |          | Revise Probability                                    | math, probability, formula, fraction style      | 113           | 2     |         |
|    | What is the Probat UVA 10056    |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, probability                               | 113           | 3     |         |
|    | HACKR sherlock-and-p            |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, probability, fractions style              | 113           | 3     |         |
|    | Probability(Given UVA 11181     |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, probability, conditional probability      | 113           | 4     | p2      |
|    | Another lottery UVA 11628       |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, probability, fraction style, gcd          | 113           | 4     |         |
|    | Mushroom Scientists CF186-D2-D  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, probability or log, ternary search        | 113           | 5     | p4      |
|    | CF101864-GYM-A                  |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, probability, combinatorics, math          | 113           | 5     | p2      |
|    | SRM537-D2-1000                  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, probability, graph, cycle                 | 113           | 5     | p2      |
|    | Airplane UVA 12461              |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, probability, greedy                       | 113           | 5     | p1      |
|    | Probability Uva 11346           |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, probability, integration                  | 113           | 5.25  | p3      |
|    | SRM285-D1-500                   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, probability, bf or dp                     | 113           | 5.5   |         |
|    | CF26-D12-D                      |        |              |                 |                  |                |               | 0             |                   |              |          | Sol - must read                                       | math, probability, factorial, logarithm, combi  | 113           | 5.5   | p3      |
|    | CF442-D1-B                      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, probability, sorting                      | 113           | 5.5   | p3      |
|    | SRM352-D2-1000                  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, probability, recursion, precision         | 113           | 5.5   |         |
|    | CF513-D12-C                     |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, probability, bitmasks or dp, probability  | 113           | 6     | p3      |
|    | UVA 557                         |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, probability, combinatorics                | 113           | 6     |         |
|    | SPOJ FUNPROB                    |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, probability, formula                      | 113           | 6     |         |
|    | CF163-D12-C                     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, probability                               | 113           | 6.25  |         |
|    | CF110-D2-D                      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, probability, combinatorics                | 113           | 6.25  |         |
|    | God, Save me UVA 10777          |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, probability, expectation or dp, probab    | 114           | 4     |         |
|    | SRM458-D2-500                   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, probability, expectation, bitmasks        | 114           | 4     |         |
|    | CF839-D2-C                      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, probability, expectation, dfs             | 114           | 4     |         |
|    | HACKR lazy-sorting              |        |              |                 |                  |                |               | 0             |                   |              |          | Revise Expected Value                                 | math, probability, expectation, permutation     | 114           | 4     |         |
|    | Andrey and Problem CF443-D2-D   |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, probability, expectation, greedy or dp    | 114           | 4.5   | p3      |
|    | Wet Shark and Flo CF621-D2-C    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, probability, expectation                  | 114           | 4.5   |         |
|    | Little Pony and Ex CF454-D2-C   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, probability, expectation, pattern         | 114           | 4.5   |         |
|    | HACKR vertical-sticks           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, probability, expectation, linearity of ex | 114           | 5     | p3      |
|    | SRM577-D1-250                   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, probability, expectation, linearity of ex | 114           | 5     | p3      |
|    | SRM470-D1-500                   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, probability, expectation                  | 114           | 5.5   | p2      |
|    | CF500-D12-D                     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, probability, expectation, dfs             | 114           | 5.5   | p2      |
|    | CF280-D1-C                      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, probability, expectation, dfs or dp       | 114           | 6     | p3      |
|    | Playlist CF268-D2-E             |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, probability, expectation, formula, gre    | 114           | 6     | p3      |
|    | Big Mod UVA 374                 |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, repeated squaring, mod, direct            | 115           | 3     |         |
|    | Twin Primes UVA 10394           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, sieve                                     | 117           | 3     |         |
|    | Factorial Factors UVA 884       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, sieve, factorization                      | 117           | 3.5   |         |
|    | Psycho SPOJ PSYCHON             |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, sieve, factorization, tricky big # test c | 117           | 4     | p2      |
|    | Summation of Fou UVA 10168 [5]  |        |              |                 |                  |                |               | 0             |                   |              |          | Video Solution - Eng Moaz Rashad [6]                  | math, sieve                                     | 117           | 4     |         |
|    | Primes or Palindrom CF569-D2-C  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, sieve, palindromes                        | 117           | 4.5   | p3      |
|    | Divisibility of Facto UVA 10484 |        |              |                 |                  |                |               | 0             |                   |              |          | Sol to read                                           | math, sieve                                     | 117           | 4.5   | p2      |
|    | LIVEARCHIVE 4008                |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, sieve, [last non zero digit of permutat   | 117           | 5.5   | p2      |
|    | The New Rule in E UVA 10742     |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, sieve, binary search                      | 117           | 5.5   |         |
|    | Sum-up the Prime UVA 10419      |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | math, sieve, dfs, dp                            | 117           | 5.5   |         |
|    | Flying Saucer Seg CF227-D2-C    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, summations                                | 118           | 4.5   | p3      |
|    | Dreamoon and Su CF476-D2-C      |        |              |                 |                  |                |               | 0             |                   |              |          | Video Solution - Dr Mostafa Saad                      | math, summations, [in my videos]                | 118           | 5     | p3      |
|    | CF201-D1-B                      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, summations, separate summations o         | 118           | 5     | p2      |
|    | Spongebob and S CF599-D2-D      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | math, summations, bf, [overflow]                | 118           | 6     | p2      |
|    | Largest Rectangle SPOJ HISTOGRA |        |              |                 |                  |                |               | 0             |                   |              |          | Sol_Don't implement as adhoc/greedy/Pure ST           | rmq, d&c or datastructure, [largest rectangl    | 122           | 4.5   | p4      |
|    | R2D2 and Droid A CF514-D2-D     |        |              |                 |                  |                |               | 0             |                   |              |          | Use rmq                                               | rmq, binary search or bit or two pointers       | 122           | 5     | p3      |
|    | Friends and Subst CF689-D2-D    |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | rmq, sparse table, binary search or datastru    | 122           | 5     | p3      |
|    | Pair of Numbers CF359-D2-D      |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | rmq, binary search, gcd, analysis or stack      | 122           | 5.5   | p2      |
|    | Square Subsets CF448-D2-C       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | search, d&c, greedy                             | 123           | 4.5   |         |
|    | Potentiometers LIVEARCHIVE 2191 |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | segment tree, [interval sum query]              | 125           | 2     | p3      |
|    | Interval Product UVA 12532      |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | segment tree or bit, [==tju 3440]               | 125           | 2     |         |
|    | Halt The War SPOJ CDC12_H       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | segment tree                                    | 125           | 3.5   |         |
|    | Multiples of 3 SPOJ MULTQ3      |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | segment tree, lazy propagation                  | 125           | 4     | p3      |
|    | Horrible Queries SPOJ HORRIBLE  |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | segment tree, lazy propagation or bit           | 125           | 4     | p1      |
|    | Counting Primes SPOJ CNTPRIME   |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | segment tree, sieve                             | 125           | 4     | p1      |
|    | Maximum Sum SPOJ KGSS           |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | segment tree, [max pair sum]                    | 125           | 4.5   | p3      |
|    | A Famous City SPOJ CITY2        |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | segment tree or adhoc                           | 125           | 4.5   | p2      |
|    | Help R2-D2! SPOJ HELPR2D2       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | segment tree, impl                              | 125           | 4.5   | p2      |
|    | Light Switching SPOJ LITE       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | segment tree, lazy propagation, [edu]           | 125           | 4.5   | p1      |
|    | Circular RMQ CF52-D12-C         |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | segment tree, lazy propagation, circular        | 125           | 4.5   |         |
|    | Brackets SPOJ BRCKTS            |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | segment tree, [bracket balance, 2 values in     | 125           | 5     | p3      |
|    | Can you answer th SPOJ GSS1     |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | segment tree, [max sum, part of gss series      | 125           | 5     | p3      |
|    | RMQ with Shifts UVA 12299       |        |              |                 |                  |                |               | 0             |                   |              |          | See sscanf and sprint usage                           | segment tree, rmq shift                         | 125           | 5     | p3      |
|    | AND Rounds SPOJ ANDROUND        |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | segment tree                                    | 125           | 5     | p2      |
|    | Ahoy, Pirates! UVA 11402        |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | segment tree, lazy propagation or datastruc     | 125           | 5     | p2      |
|    | Present CF460-D2-C              |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | segment tree, lazy propagation, greedy or t     | 125           | 5     | p2      |
|    | Fence Obstacle C PKU 2374       |        |              |                 |                  |                |               | 0             |                   |              |          | Sol                                                   | segment tree, dp or dp                          | 125           | 5     | p1      |
|    | CF61-D2-E                       |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | segment tree or wavelet tree, [boring, inver    | 125           | 5     | p1      |
|    | Can you answer th SPOJ GSS3     |        |              |                 |                  |                |               | 0             |                   |              |          |                                                       | segment tree, [max sum+updates, spoj gss        | 125           | 5.5   | p5      |





| ff | Problem Code   | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level /10 | By yourself? | Category | 1-2 line Comments about your approach is interesting?                          | Mostafa Category | Category Code | Level | Quality |
|----|----------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|-------------------|--------------|----------|--------------------------------------------------------------------------------|------------------|---------------|-------|---------|
|    | AC Averages => | 0      | 0            | 0               | 0                | 0              | 0             | 0             | 0                 | 0            | 0        | 0                                                                              |                  |               |       |         |
|    |                |        |              |                 |                  | 62             |               |               |                   | 63           |          | <a href="#">Watch - Graph Theory - Tree Diameter and Isomorphism</a>           |                  |               |       |         |
|    |                |        |              |                 |                  | 114            |               |               |                   | 64           |          | <a href="#">Watch Video - Expected Value</a>                                   |                  |               |       |         |
|    |                |        |              |                 |                  | 122, 125       |               |               |                   | 65           |          | <a href="#">Watch - Data Structures - Segment Tree (2 vid)</a>                 |                  |               |       |         |
|    |                |        |              |                 |                  | 38             |               |               |                   | 66           |          | <a href="#">Reading: DP on Trees</a>                                           |                  |               |       |         |
|    |                |        |              |                 |                  | 138            |               |               |                   | 67           |          | <a href="#">Watch - Two pointers technique</a>                                 |                  |               |       |         |
|    |                |        |              |                 |                  | 29             |               |               |                   | 68           |          | <a href="#">Watch - DP - Probability</a>                                       |                  |               |       |         |
|    |                |        |              |                 |                  | 11             |               |               |                   | 69           |          | <a href="#">Watch - DP - Masks (2 vid)</a>                                     |                  |               |       |         |
|    |                |        |              |                 |                  | 135            |               |               |                   | 70           |          | <a href="#">Watch - String Processing - Trie</a>                               |                  |               |       |         |
|    |                |        |              |                 |                  | 36             |               |               |                   | 71           |          | <a href="#">Watch - DP - Sub-rectangle style</a>                               |                  |               |       |         |
|    |                |        |              |                 |                  | 130            |               |               |                   | 72           |          | <a href="#">Watch - String Processing - KMP (2 vid)</a>                        |                  |               |       |         |
|    |                |        |              |                 |                  | 23             |               |               |                   | 73           |          | <a href="#">Watch - DP - Games (2 vid)</a>                                     |                  |               |       |         |
|    |                |        |              |                 |                  | 49             |               |               |                   | 74           |          | <a href="#">Watch - Computational Geometry - Simple and Convex Polygons</a>    |                  |               |       |         |
|    |                |        |              |                 |                  | 49             |               |               |                   | 75           |          | <a href="#">Watch - Computational Geometry - Polygon Area - Centroid - Cut</a> |                  |               |       |         |
|    |                |        |              |                 |                  | 49             |               |               |                   | 76           |          | <a href="#">Watch - Computational Geometry - Point in polygon</a>              |                  |               |       |         |
|    |                |        |              |                 |                  | 71,72,74       |               |               |                   | 77           |          | <a href="#">Watch - Graph Theory - Maximum Flow (2 vid)</a>                    |                  |               |       |         |
|    |                |        |              |                 |                  | 77             |               |               |                   | 78           |          | <a href="#">Watch - Graph Theory - SCC (2 vid)</a>                             |                  |               |       |         |

| Problem Name   | Problem Code | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level/10 | By yourself? | Category | 1-2 line Comments about your approach is interesting? | Mostafa Category | Category Code | Level | Quality |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|----------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|------------------|----------------|---------------|---------------|------------------|--------------|----------|-------------------------------------------------------|------------------|---------------|-------|---------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| AC Averages => |              | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0            | 0               | 0                | 0              | 0             | 0             | 0                | 0            |          | 0                                                     |                  |               |       |         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                |              | - Problems in this topic2 page <b>don't exist</b> in any other sheet (A, B, C, D, Topics1)<br>- For juniors, you should <b>ignore</b> this page till finish the previous sheets FIRST<br>- These topics represent 'most' of the other algorithms that <b>seniors</b> may learn<br>- Problems per topic cover range of easy to medium level (hard Div2-D)<br>- The purpose is to allow some of the trainees who want to expand in several algorithms before going deeper or whoever want to enhance a specific topic<br>- 50-70% of the problems here will be of right level/quality. However, many of them are not evaluated by enough (or no) trainees. More future refinements may occur based on feedback<br><br>- Help me refine this list. Consider these cases:<br>- Problem that is so bad somehow<br>- Problem of 'p3, p4, p5' quality but actually is a repeated idea or not interesting<br>- Problem of 'empty, p1, p2' quality but actually is very interesting to solve<br>- Problem of really wrong level (so easy/so hard) relative to my reported level<br><br>If any of these issues, email me: mostafa.saad.fc@gmail.com<br><br>Also, if a problem doesn't have an editorial and you could provide it = great! |              |                 |                  |                |               |               |                  |              |          |                                                       |                  |               |       |         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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| Problem Name                                                                                                              | Problem Code                                                                                                         | Status | Submit Count | Reading Time(m) | Thinking Time(m) | Coding Time(m) | Debug Time(m) | Total Time(m) | Problem Level/10 | By yourself? | Category | 1-2 line Comments about your approach is interesting? | Mostafa Category                                                                  | Category Code     | Level          | Quality |
|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------|--------------|-----------------|------------------|----------------|---------------|---------------|------------------|--------------|----------|-------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------|----------------|---------|
|                                                                                                                           | AC Averages =><br><a href="#">HACKER uchiha-broth</a><br><a href="#">CF632-D12-E</a><br><a href="#">SPOJ POLYMUL</a> |        | 0            | 0               | 0                | 0              | 0             | 0             | 0                | 0            | 0        | 0                                                     | dp, dp_sos, inclusion-exclusion<br>math, fft, [practice]<br>math, fft, [practice] | 134<br>138<br>138 | 6.25<br>6<br>6 | p4      |
| In case you want some (subjective) order to study these topics:<br><a href="#">This color means covered in my channel</a> |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">geometry, sweep line</a>                                                                                      |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">graph, lca</a>                                                                                                |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">dp, dp_convex_hull</a>                                                                                        |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">mo's Algorithm</a>                                                                                            |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">sqrt decomposition</a>                                                                                        |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">game theory, nim</a>                                                                                          |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">game theory, grundy</a>                                                                                       |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">string processing, polynomial hashing</a>                                                                     |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">graph, min-cost-max-flow</a>                                                                                  |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">ternary search</a>                                                                                            |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">binary indexed tree</a>                                                                                       |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">math, extended gcd</a>                                                                                        |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">math, mod inv</a>                                                                                             |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">graph, scc, articulation point</a>                                                                            |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">graph, scc, biconnected components</a>                                                                        |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">math, matrix, matrix pow</a>                                                                                  |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">graph, euler tour</a>                                                                                         |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">dp, dp_d&amp;c_opt</a>                                                                                        |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">dp, dp_expectation</a>                                                                                        |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">graph, bellmanford</a>                                                                                        |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">meet in middle</a>                                                                                            |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">graph, heavy light decomposition</a>                                                                          |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">graph, dsu-on-trees</a>                                                                                       |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">graph, centroid-decomposition</a>                                                                             |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">grid compress</a>                                                                                             |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">math, matrix, gaussian elimination</a>                                                                        |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">dp, dp_knuth</a>                                                                                              |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">dp, dp_state_reduce</a>                                                                                       |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">dp, dp_sos</a>                                                                                                |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">graph, scc, 2-sat</a>                                                                                         |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">math, cyclic permutation</a>                                                                                  |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">string processing, suffix array</a>                                                                           |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">balanced binary search tree</a>                                                                               |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">math, diophantine</a>                                                                                         |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">math, totient</a>                                                                                             |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">math, mobius</a>                                                                                              |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">string processing, aho_corasick</a>                                                                           |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">geometry, polygon, convex hull</a>                                                                            |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">math, fft</a>                                                                                                 |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">dp, dp_cyclic</a>                                                                                             |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |
| <a href="#">geometry, 3d</a>                                                                                              |                                                                                                                      |        |              |                 |                  |                |               |               |                  |              |          |                                                       |                                                                                   |                   |                |         |

|                      | Weekly Check List                                                                                                                    |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
|                      | <a href="#">Training Secrets of Success Video</a>                                                                                    |
|                      |                                                                                                                                      |
| Reading              |                                                                                                                                      |
|                      | Read within 3-5 minutes for short text problem. If no, you need to work on your Reading English Skills                               |
|                      | Never suspect later your problem understanding? If happens, you need to improve your comprehension / cases tracing                   |
|                      |                                                                                                                                      |
| Thinking             |                                                                                                                                      |
|                      | Ready and in the challenging mood before start solving.                                                                              |
|                      | <a href="#">Striving against the problem for a reasonable time. If no, you need to change your solving spirit. Be a fighter.</a>     |
|                      | Found a solution; Do verifications: text cases / extra cases / correctness / time & memory                                           |
|                      |                                                                                                                                      |
| Coding               |                                                                                                                                      |
|                      | Sketch in your mind the big picture of the code first. Don't rush for coding                                                         |
|                      | Code within 10 minutes. If more, you have coding skills problem or your understanding for the approach is not complete               |
|                      | A lot of copy paste? Something wrong. Need a better code organization                                                                |
|                      | Needed more than 10 minutes to code medium size codes? Why? Identify the issue and solve it                                          |
|                      |                                                                                                                                      |
| Debugging            |                                                                                                                                      |
|                      | Which will be faster to catch the mistake? Printing or Debugger                                                                      |
|                      | Don't know how to use a debugger? Learn this skill                                                                                   |
|                      | Needed more than 10 minutes to solve bugs? Something is wrong. Why need all this time? How to solve this issue?                      |
|                      |                                                                                                                                      |
| Code is ready!       |                                                                                                                                      |
|                      | Just submit and see if passed? Wrong. Behave as if you are in the real contest. Are you almost sure it will be AC? If yes, submit    |
|                      | TRAIN offline as if you are in a real contest. This shortens the gap between training and the real contest                           |
|                      |                                                                                                                                      |
| Code Failed :(       |                                                                                                                                      |
|                      | Are you nervous / frustrated? Yes => Wrong behavior. Take it easy                                                                    |
|                      | Rush to test cases? Yes => Wrong, revise idea, then code, then trace more samples. Try for 15 minutes or more first                  |
|                      |                                                                                                                                      |
| Got it AC            |                                                                                                                                      |
|                      | Read and Understood editorial solutions?                                                                                             |
|                      | Checked 1-3 other AC solutions?                                                                                                      |
|                      | Tried to write a much shorter version of your code?                                                                                  |
|                      | Tried to write a faster coder (better complexity)?                                                                                   |
|                      |                                                                                                                                      |
| Speed                |                                                                                                                                      |
|                      | How much time do you need in Div2-A/Div2-B? Target (5, 10) minutes for semiseniors, (3, 6) for seniors                               |
|                      | Not that fast? You need regular speed training on easy problems                                                                      |
|                      |                                                                                                                                      |
| Weakly contests      |                                                                                                                                      |
|                      | Do you participate in 1-2 contests per week at least? If no, this is bad. Offline training != Online contests                        |
|                      | You need to train yourself to behave in online contests similar to offline training. This is an important skill.                     |
|                      |                                                                                                                                      |
| Sheet stats          |                                                                                                                                      |
|                      | Recorded them? Yes: read your problem's row. Where do you consume the most of the time? These are your weak skills                   |
|                      | No, I don't record! => How will you know your weak points?!                                                                          |
|                      | Can't record timing as I am mixing thinking with coding? => Wrong behavior. get done with thinking, then move to coding. Don't cycle |
|                      |                                                                                                                                      |
| Training Time        |                                                                                                                                      |
|                      | Is it regular and scheduled? Yes => you will have regular improvements.                                                              |
|                      | Your plan was to train X hours, Did so? If no, why?                                                                                  |
|                      | Without regular and continous training, your mind might not improve well                                                             |
|                      |                                                                                                                                      |
| Training with?       |                                                                                                                                      |
|                      | Yourself only? You may feel bored. If can collaborate with others = longer commitment                                                |
|                      |                                                                                                                                      |
| Psychological issues |                                                                                                                                      |
|                      | Do you keep comparing yourself with others?                                                                                          |
|                      | Do you have negative feelings? Like I am stupid...I am hopeless...I will never have a comparable level...?                           |
|                      | Do you think of your image/appearance if failed in online contests so avoid contests?                                                |
|                      | Do you use another account with a weird name to train so that people don't know about your progress/failure?                         |
|                      | Do you wish your friends fail in the contest? or get annoyed with their better performance?                                          |
|                      | Do you avoid teaching your friends something or give no support to remain better than them?                                          |

|  |                                                                                                      |
|--|------------------------------------------------------------------------------------------------------|
|  | Do you feel bored/frustrated as no/weak community in your college?                                   |
|  | Do you keep training day and night without breaks? No socialization at all?                          |
|  | Do you hate specific topics and avoid them (probability/geometry)?                                   |
|  | 'Should I stop' Dilemma? Keep thinking is it worth vs a waste of time?                               |
|  |                                                                                                      |
|  | If any of the above questions is YES, you probably have a problem and need to find a solution to it. |



[illegible]

[illegible]

[1] Google first Goldbach's conjecture

[2] Or shorter:

[http://naivered.github.io/2016/07/03/Problem\\_Solving/UVa/UVa-10168-Summation-of-Four-Primes/](http://naivered.github.io/2016/07/03/Problem_Solving/UVa/UVa-10168-Summation-of-Four-Primes/)

[3] Or shorter:

<https://github.com/ilyesLtifi/Competitive-Programming/blob/master/UVa/UVa%2010325.cpp>

[4] Or shorter:

<https://github.com/ilyesLtifi/Competitive-Programming/blob/master/UVa/UVa%2010325.cpp>

[5] Google first Goldbach's conjecture

[6] Or shorter: [http://naivered.github.io/2016/07/03/Problem\\_Solving/UVa/UVa-10168-Summation-of-Four-Primes/](http://naivered.github.io/2016/07/03/Problem_Solving/UVa/UVa-10168-Summation-of-Four-Primes/)

[7] From a trainee: txt clarification: In this problem FJ can actually REARRANGE the plots and then groups them. \* I misunderstood the statement by the word 'successive'